

RESTRICTED

BINDING DATA

FOR

**MAIN BROADCAST / SOUND REPRODUCING EQUIPMENT
(MB/SRE) SYSTEM**

FOR

DSV VC 11190-91

Drawing Ref. No. : BD/P1263/MB
Issue No. : 00
Customer Name : HINDUSTAN SHIPYARD LIMITED
PO No : 4500001687 DATED 30.04.2021

Manufacturer's Details:

M/s Phi Audiocom Systems Pvt. Ltd.
C-10, 11(13A), BG.BLOCK
BHOSARI MIDC,
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Date: 19 JUL 2021

RESTRICTED

CERTIFICATE

**This is to certify that the technical
contents in the**

**‘BINDING DATA
FOR
MAIN BROADCAST / SOUND
REPRODUCING EQUIPMENT (MB/SRE)
SYSTEM ’**

**are complete and correct to the best of
our knowledge.**



Rajiv Khot
Project Manager
020-67306721

SAFETY WARNING

The voltage employed in the equipment
are sufficiently high to endanger human life

**POWER
MUST BE SWITCHED OFF**
before servicing the equipment

and

GREAT CARE
taken when making internal adjustments etc.

FOR FIRST AID IN CASE OF ELECTRIC SHOCK SEE OVERLEAF

FIRST AID IN CASE OF ELECTRIC SHOCK

FIG. 1



FIG. 2



FIG. 3



FIG. 4



1. SWITCH OFF: If this is not possible, PROTECT YOURSELF with dry insulating material & pull the victim clear of the conductor but DO NOT TOUCH THE VICTIM WITH YOUR BARE HANDS until he is clear of the conductor but DO NOT WASTE TIME.

2. (a) Place the victim in a supine position.
(b) Keep the air passages clear by turning the head to one side, opening the patients mouth & cleaning it of water, saline, mucus or blood, a lot of which might have accumulated in the back of the throat (Fig. 1)

3. If the jaw is rigid, try to force the mouth open by pressure on the gum behind the last molar tooth of the lower jaw. When the upper air passages are thus cleared, hit the head backward & force the jaw forward from the angles of the jaw in front of the ears. This would prevent mechanical obstructions to the upper air passages (Fig. 2, 3).

4. (a) Then hold the chin up & forward with one hand & pinch the nostrils of the victim with the other (Fig. 4).
(b) Take a very deep breath & apply your mouth to that of the victim & blow into his mouth, until the chest of the victim moves up indicating filling of the lungs (NEVER ALLOW THE CHIN TO SAG) (Fig. 4).

5. When the chest has moved up, withdraw your mouth & allow the chest to sink back. REPEAT this process every three to four seconds until the victim begins to breathe again or until he is taken over by a medical attendant. This method can be continued in an ambulance during transit of the patient from the site of accident to the nearest medical center.

HAVE SOMEONE ELSE SEND FOR A DOCTOR.
KEEP PATIENT WARM & LOOSEN HIS CLOTHING.
DO NOT GIVE LIQUIDS UNTIL THE PATIENT IS CONSCIOUS.

CAUTION / NOTES

1. THE EQUIPMENT IN THIS SYSTEM INVOLVES HIGH VOLTAGES. A DETAILED STUDY OF USER HANDBOOK AND TECHNICAL MANUAL SHOULD BE DONE PRIOR TO USING THE EQUIPMENT. FAILURE TO DO SO MAY RESULT IN THE FOLLOWING:
 - A. ELECTRIC SHOCK HAZARD FOR USER LEADING TO UNSAFE OPERATING CONDITION AND INABILITY ON THE PART OF THE OPERATOR TO OBTAIN THE DESIRED PERFORMANCE OF THE EQUIPMENT TO THE SPECIFICATION LISTED.
 - B. DAMAGE TO CERTAIN PARTS OF THE EQUIPMENT DUE TO WRONG USE
2. WHILE REPLACING GLASS FUSE, HOLDER OR ON/OFF SWITCH, IT SHOULD BE STRICTLY OBSERVED THAT THE MAINS CORD PLUG IS REMOVED FROM THE POWER BOARD.
3. ALWAYS USE 3 PIN POWER BOARD WITH PROPER EARTHING.
4. DO NOT REMOVE THE COVER OF THE EQUIPMENT AND ATTEMPT REPLACEMENT OF COMPONENTS WITHOUT REMOVING THE MAINS PLUG FROM THE MAIN SOCKET.
5. SERVICING BY USER SHOULD ONLY BE TO REPLACEMENT OF FUSE.
6. FAULT LOCATION OR REPAIRS SHOULD ONLY BE ATTEMPTED BY TRAINED PERSONNEL WITH ADEQUATE TEST EQUIPMENT AND TOOLS OBSERVING ALL THE PRECAUTIONS DETAILED IN THE TECHNICAL/SERVICE MANUAL. FAILURE TO DO SO MAY DAMAGE SEMICONDUCTORS IC'S AND PCB TRACKS.

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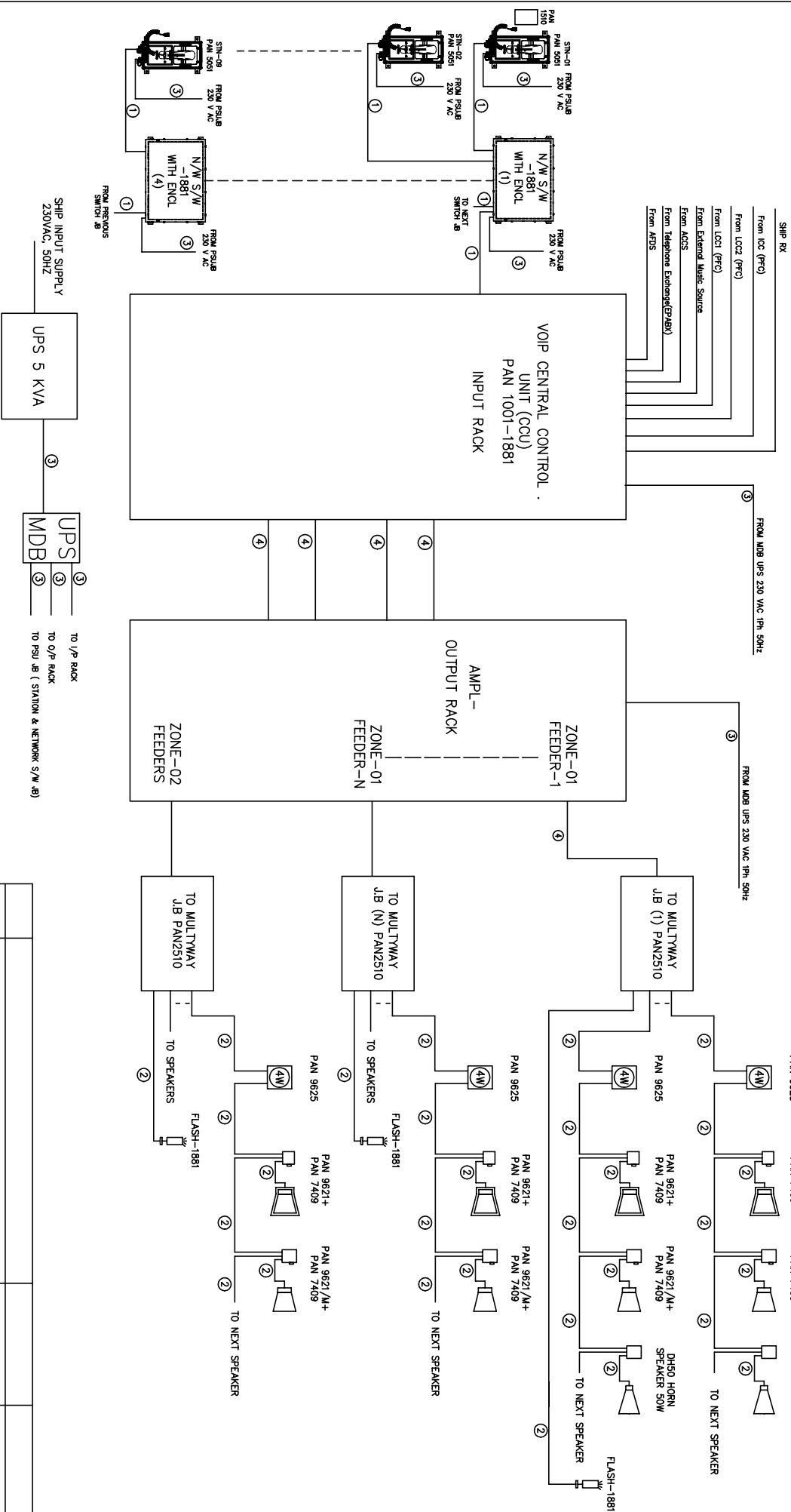
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APPROVED BY: SMK		DATE: 14/07/2021					
M/s PHI AUDIOCOM SYSTEMS PVT LTD		LIST OF CONTENTS FOR MB/SRE SYSTEM					
		DOC REF. NO.: BD/P1263/MB			SHEET	01	OF

1. BILL OF MATERIAL
PER SHIP

S. No.	Item Description	Qty.	Unit
A	Main Broadcast/ Sound Reproducing Equipments (MB/ SRE); Material No- 1200002523 consist of following:-	1	SET
1	Main unit (Central control unit with Single server, DVD Player, UPS, Control circuitry & Amplifier Rack)	1	Set
2	Box Loudspeaker 4W(NWT) with Volume Control; PAN9625	172	Nos
3	Horn Loudspeaker (15W) + Volume Control Unit; PAN9621 + PAN7409	110	Nos
4	Horn Loudspeaker (Blastproof & Dustproof); PAN9621/M + PAN7409	8	Nos
5	Horn Speaker 50W; DH50.	2	Nos
6	Wall Mounted Handset Station; PAN5051	9	Nos
7	Handset with Plug; PAN8512	9	Nos
8	Flasher (COTS Item); FLASH-1881	12	Nos
9	Multiway Junction Box; PAN2510	7	Nos
10	Network Switch 8 Port with Enclosure; SWITCH-1881	4	Nos
11	Stowage Box for Mic/ Handset; PAN1510	5	Nos
12	UPS; UPS-1881	1	No
13	Batteries for 1Hr back-up ; BATT-1881	1	Set
14	Installation and Commissioning Material	1	Set
15	On board spares	1	Set
16	Drawing & Documentation	9	Set

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	DOC REF. NO.: BD/P1263/MB	SHEET	01	OF

TYPICAL CABLING SCHEMATIC DIAGRAM
FOR MB/SRE SYSTEM: SH-01



REQUIRED CABLE TYPES :-

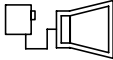
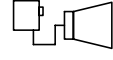
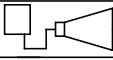
- ① CAT6 SHIELDED CABLE (LAN CABLE)
- ② 4X0.75 SQ.MM CABLE (LOUDSPEAKER CABLE)
- ③ 2CX2.5SQ.MM CABLE (MAINS CABLE)
- ④ 10/20X0.75 CABLE (INTEGRAL SIGNAL CABLE)

NOTE: 1) ALL THE OFFERED SPEAKERS ARE ANALOG TYPE. OFFERED SYSTEM IS SAME AS P80

- 2) OFFERED JBS ARE AS PER HSL REQUIREMENT AND SUPPLIED AGAINST TC-GC
- 3) FORM CENTRAL CONTROL UNIT TO N/W S/W JB: 70 TO 90 MTRS.
- 4) N/W S/W JB TO N/W S/W JB: 70 TO 90 MTRS.
- 5) N/W S/W JB TO FIELD STATION: 50 MTRS.
- 6) MAX 04 NOS OF FIELD CALL STATIONS CONNECTED TO EACH NETWORK SWITCH JB.

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G.A.		DRAWING REF. NO.: BD/P1263/MB		SHEET	01 OF 02

TYPICAL CABLING SCHEMATIC DIAGRAM FOR MB/SRE SYSTEM LEGENDS: SH-02

Eqpt No.	Symbol	Description	Qty
1		Box Loudspeaker 4W(NWT) with Volume Control; PAN9625	172 NOS
2		Horn Loudspeaker (15W) + Volume Control Unit; PAN9621 + PAN7409	110 NOS
3		Horn Loudspeaker (Blastproof & Dustproof); PAN9621/M + PAN7409	8 NOS
4		Horn Speaker 50W; DH50	2 NOS
5		Wall Mounted Handset Station; PAN5051	9 NOS
6		Handset with Plug; PAN8512	9 NOS
7		Flasher (COTS Item); FLASH-1881	12 NOS
8		Network Switch 8 Port with Enclosure; SWITCH-1881	4 NOS
9		CENTRAL CONTROL UNIT	1 SET
10		AMPLIFIER RACK	
11		Multiway Junction Box; PAN2510	7 NOS
12		Stowage Box for Mic/ Handset; PAN1510	5 NOS
13		UPS; UPS-1881	1 NO

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	CHECKED BY : RKK	DATE : 09/07/2021		
	APPROVED BY : SMK	DATE : 12/07/2021		
M/S PHI AUDIOCOM SYSTEMS PVT. LTD.		TYPICAL CABLING SCHEMATIC DIAGRAM FOR MB/ SRE SYSTEM		
		G.A.	DRAWING REF. NO.: BD/P1263/MB	SHEET 02 OF 02

SYSTEM DESCRIPTION OF MB/SRE SYSTEM

1. INTRODUCTION:-

MB/ SRE system is designed to fulfill internal communication needs of Indian Naval Ships, Viz All call, Alarm call and Zone call communication. It is basically a IP based system. It has various programmable and diagnostics features through GUI (Graphical User Interface).

System is fully configurable on board without changing any hardware and cabling. System works on 230V AC, 1 Phase supply.

2. SYSTEM ELEMENTS:-

Major system elements are as follows:

2.1 MAIN UNIT (CENTRAL CONTROL UNITS);

Main Unit is used for System configuration & diagnostics through GUI, signal processing, signal controlling, taking inputs / giving outputs to / from stations, Speakers etc.

Central Control Unit is MS structure Rack. It typically consists of 19” Audio Input/Output Subrack, Server, Network Switch, Music Source, Amplifiers & other accessories like Mains Panel, Fans, Cable Termination arrangement, etc as per Ship requirement. For cable entry, detachable Gland Plate is provided on Top side of Central Control Unit.

(a) SERVER: - Server is a main controlling part of system. Server controls all the functionality of the system. All the features of the system (i.e. point to point, group, global & alarm call) will be configured through server using GUI.

(b) NETWORK SWITCH :- Single Ethernet Switch is used to Trans-receive the data. Internally network switch is connected to server, Audio Interface Module (AIM) & Zone Module. Central Control Unit Network switch is connected to Field Station through Network Junction box.

(c) AUDIO INPUT MODULES (AIM):- One Audio Input module is use for interface with the Internal/ external Audio (Music) system. Play music on music player/ Audio system, the music gets relayed on speakers.

(d) ZONE MODULES:- Zone modules are used to provide audio output to MB/SRE system (Main broadcast announcements / alarms in MB/SRE system).

(e) MUSIC SOURCE:- System has inbuilt Music Source. For sound entertainment, Sony make music source, tuner radio with FM/SW/MW/CD/DVD/MP3/USB player has been used.

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2.2 WALL MOUNTED HANDSET STATION (PAN 5051) ;

This is a wall mounting Type station suitable for indoor applications. It is to be used in conjunction with Handset. Cradle is provided on unit itself for resting the Handset. Through GUI, this station can be configured for initiating calls (Alarms, All/Global and Zone) as per Ship requirements. Keyboard with multiple keys are provided which can be used for different functionality. Wall mounted Handset station is connected to Junction box (Network switch) through CAT6 Ethernet cable.

2.3 NETWORK SWITCH WITH ENCLOSURE (SWITCH-1881);

This is wall mounting Junction Box. It is to be used for interlink between Central control unit and Wall Mounted Handset station. The Network Switch Junction Box is connected to Central Control Unit through CAT6 cable & the Wall Mounted Handset Station is connected to Network Switch JB through CAT6 Cable.

2.4 OUTPUT OR DISTRIBUTION UNITS: - These units consists of various types of Non-Watertight loudspeakers (PAN9625 – Set at 1W or 4W with volume control) suitable for below deck applications, Watertight loudspeakers (PAN9621+PAN7409 – Set at 4W/8W/15W) suitable for above deck applications and blast-proof & dust-proof loudspeakers (PAN9621/M+PAN7409 – Set at 4W/8W/15W) suitable as per ship requirements. For watertight Loudspeakers, PAN9621 / PAN9621/M are Horns and PAN7409 are volume control units. Junction Boxes Type (PAN2510) are used for cable distribution purpose.

3. CALL DETAILS:-

In this system followings Announcements, Alarms, Communication calls are possible. The system can be configured to support following types of calls:-

- i. Alarm Call
- ii. Global/ All Call
- iii. Zone Call
- iv. Music

(A) ONE WAY MB CALLS:

3.1 Alarm Call: - When station initiates Alarm using direct call key, then alarm gets relayed on All zones speakers connected to the system. Music previously relayed over overhead speakers gets disconnected and alarm gets relayed on speaker. After alarm is over i.e. after predefined Alarm Call timeout or if initiator presses Reset Key then music gets resume. Alarms can be Standard tones or prerecorded messages.

3.2 Global/ All Call: - When station dials a direct call key assigned for global call, then announcement of user gets relayed on all overhead speakers connected to the field call stations. Music previously relayed over overhead speakers gets disconnected and announcement gets relayed field call stations speaker. After announcement is over i.e. after predefined Call timeout or if initiator presses Reset Key then music gets resume.

3.3 Zone Call: - When station dials a direct call key assigned for Zone call (Zone-01 & Zone-02), then

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announcement of user gets relayed on all overhead speakers connected to the respective Zones. Music previously relayed over the speakers gets disconnected and announcement gets relayed on Zones speaker. After announcement is over i.e. after predefined Call timeout or if initiator presses Reset Key then music gets resume.

(B) SRE System: - Sound Reinforcement (SRE) is used for sound entertainment for personnel on-board. It uses Music Source (External or Internal); Tuner/CD/DVD/MP3/USB Player. Signal from this unit can be selected and relayed over the loudspeakers distributed in the zones. SRE can be optional locations wise i.e. it can be enabled or disabled for any particular loudspeaker (through hard wiring at the time of installation) for cabin type speakers.

4. CALL PRIORITIES:-

Alarm > All call / Broadcast Call > Zone call > Music.

5. DIAGNOSTIC FEATURES (Monitoring of System elements):-

System is having built-in diagnostic facility to check health of Field Call Stations at its respective Server. Fault log is maintained for references of System Administrator and Maintenance In-charge.

6. CONFIGURATION FEATURES:-

I. It is possible for the system administrator to enable or disable particular call access to each Field Call Station. Also, administrator is able to program communication chart for each Field Call Station i.e. Administrator is able to alter calling list for any Field Call Station through Graphic User Interface (GUI-Loaded on Server). Alteration of Calling List for any Field Call Stations doesn't require addition/deletion of hardware in the system.

II. During installation and commissioning of the system administrator will decide the kind of facility to be provided for each Field Call Station and it is possible to enable/disable the desired facilities through GUI software. Following functions for each Field Call Station can be programmed through GUI.

- a) Station No.
 1. Priority Levels.
 2. Global Call facility.
 3. Global Alarm Call Facility.
 4. Zone call facility
 5. Direct Call keys programming.

7. ENVIROMENTAL SPECIFICATIONS:-

All units meet following environmental specifications.

(a) Operating Temperature : - 40° C to +55° C.

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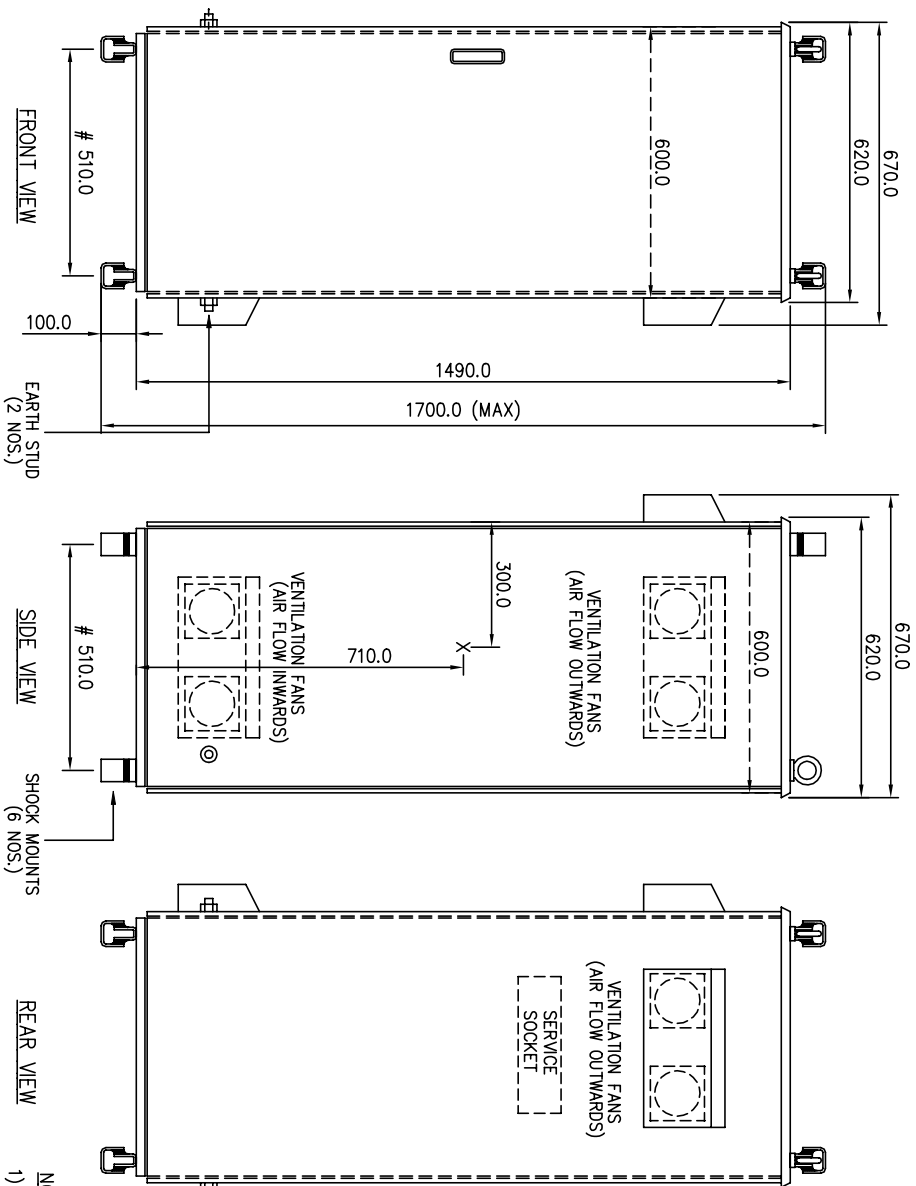
(b) Relative Humidity : 0 - 95% RH condensing at 40° C.

Note: The maximum cable distance of the equipment's are mentioned as follows:

- i. CENTRAL CONTROL UNIT TO NETWORK SWITCH JB: 70 TO 90 MTRS. (CAT 6 cable)
- ii. NETWORK SWITCH JB TO NETWORK SWITCH JB: 70 TO 90 MTRS. (CAT 6 cable)
- iii. NETWORK SWITCH JB TO WALL MOUNTED HANDSET STATION: 50 MTRS. (CAT 6 cable)

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**GENERAL ARRANGEMENT DRAWING OF
CENTRAL CONTROL UNIT; PAN1001-1881-1**



- NOTE:-**
- 1) MOUNTING : FLOOR MOUNTING.
 - 2) FINISH : DA GREY, COLOUR CODE-632
 - 3) HARDWARE USED FOR ASSEMBLY IS STANDARD STAINLESS STEEL INDUSTRIAL GRADE HARDWARE.
 - 4) OPERATING VOLTAGE : 230V, 1 ϕ , 50 HZ, AC POWER SUPPLY.
 - 5) POWER CONSUMPTION : 1000VA MAX.
 - 6) HEAT DISSIPATION : 500W MAX.

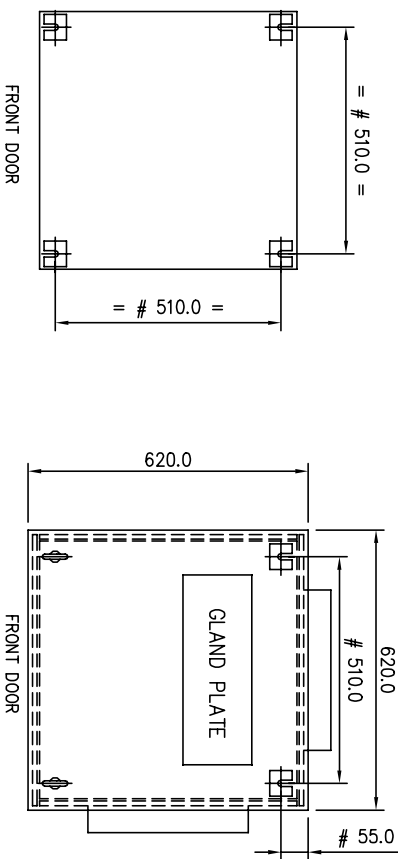
WEIGHT - 200 KGS MAX
ALL DIMENSIONS IN MM, GENERAL TOLERANCE ± 10 MM,
MOUNTING DIMENSION, TOLERANCE ± 2 MM.

NOTE :

- 1) ALL DIMENSIONS ARE IN MM, GEN. TOLERANCE = ± 10 mm
- 2) MINIMUM DISTANCE OF RACK FROM WALL ON FRONT & REAR SIDE SHOULD BE 700MM
- 3) ACTUAL POSITION MAY CHANGE ACCORDING TO AVAILABILITY AND SUITABILITY OF SPACE.

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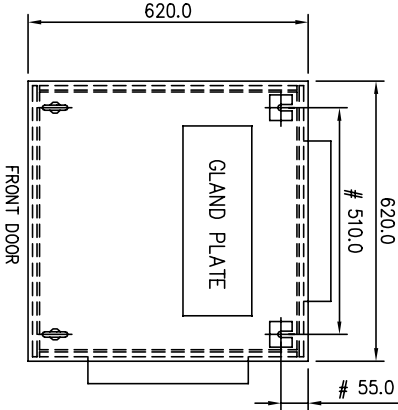
1. REFER SHEET 02 FOR TECHNICAL SPECIFICATION.



RACK MOUNTING DETAILS (BOTTOM VIEW)

MOUNTING BOLTS M16-4x
TO BE LOCATED AT THESE DIMENSIONS.

TOP VIEW (RACK MOUNTING DETAILS)



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M/S PHI AUDIOCOM
SYSTEMS PVT. LTD.

CENTRAL CONTROL UNIT ; MODEL:- PAN1001-1881-1

G.A. DRAWING REF. NO.: BD/P1263/MB SHEET 01 OF 04

TECHNICAL SPECIFICATIONS OF MAIN UNIT (CENTRAL CONTROL UNIT) FOR FOR MB/SRE SYSTEM (PAN1001-1881-1)

Central Control Unit is a main controlling Unit of IP Based Audio Communication Systems. It typically consists of various Types of Subracks – Control circuitry and Tray/ Panels based on the Ship's requirement. Also, configuration of various cards used in these Subracks depends on Ship requirement.

1. Input Supply voltage : 230V AC, 50Hz, single phase Mains Supply From UPS.
2. Typical VA consumption : 1000 VA Max @ speech load.
3. Inputs Interface :
i) Radio / CD / DVD / USB / Player
ii) Input from Ships Receiver (Ship Rx) .
iii) AFDS, ICC, LCC & EPABX.
4. External connections : Through screw-less Connector terminals.
5. Call Facilities :
Alarm call
Global Announcement call
Zone call
Music
6. Priorities : Global Alarm > All Call > Zone call > Music
8. Controls & indications : Mains ON indication, mains switch on mains panel & Service socket for servicing purpose on base plate
9. Dimension : 670mm (D) x 670mm (W) x 1700mm (H) Max.
10. Weight : 200.0 Kg approx.
11. Environmental specifications : JSS 55555 Test Nos. 10, 17, 20, 24 & 28.
12. EMI/EMC specifications : compliant with MIL STD 461-E.

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M/s PHI AUDIOCOM SYSTEMS PVT LTD		MAIN UNIT (CENTRAL CONTROL UNIT) FOR MB/SRE SYSTEM					
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GENERAL ARRANGEMENT DRAWING FOR

MAIN UNIT (AMPLIFIER RACK): PAN1001-1881-2

TECHNICAL SPECIFICATION :-

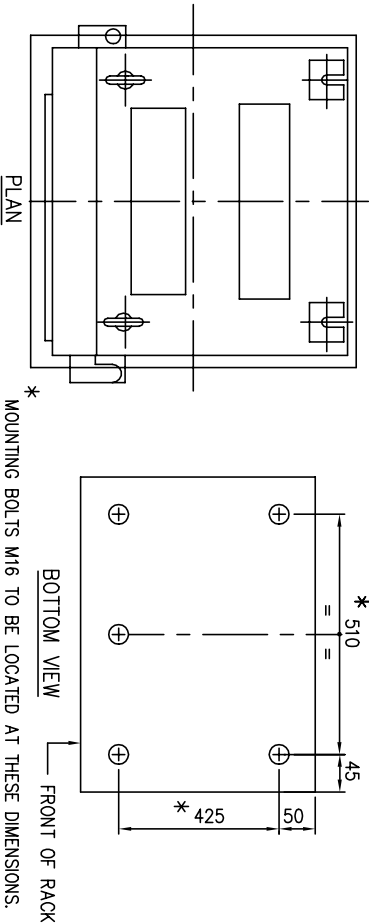
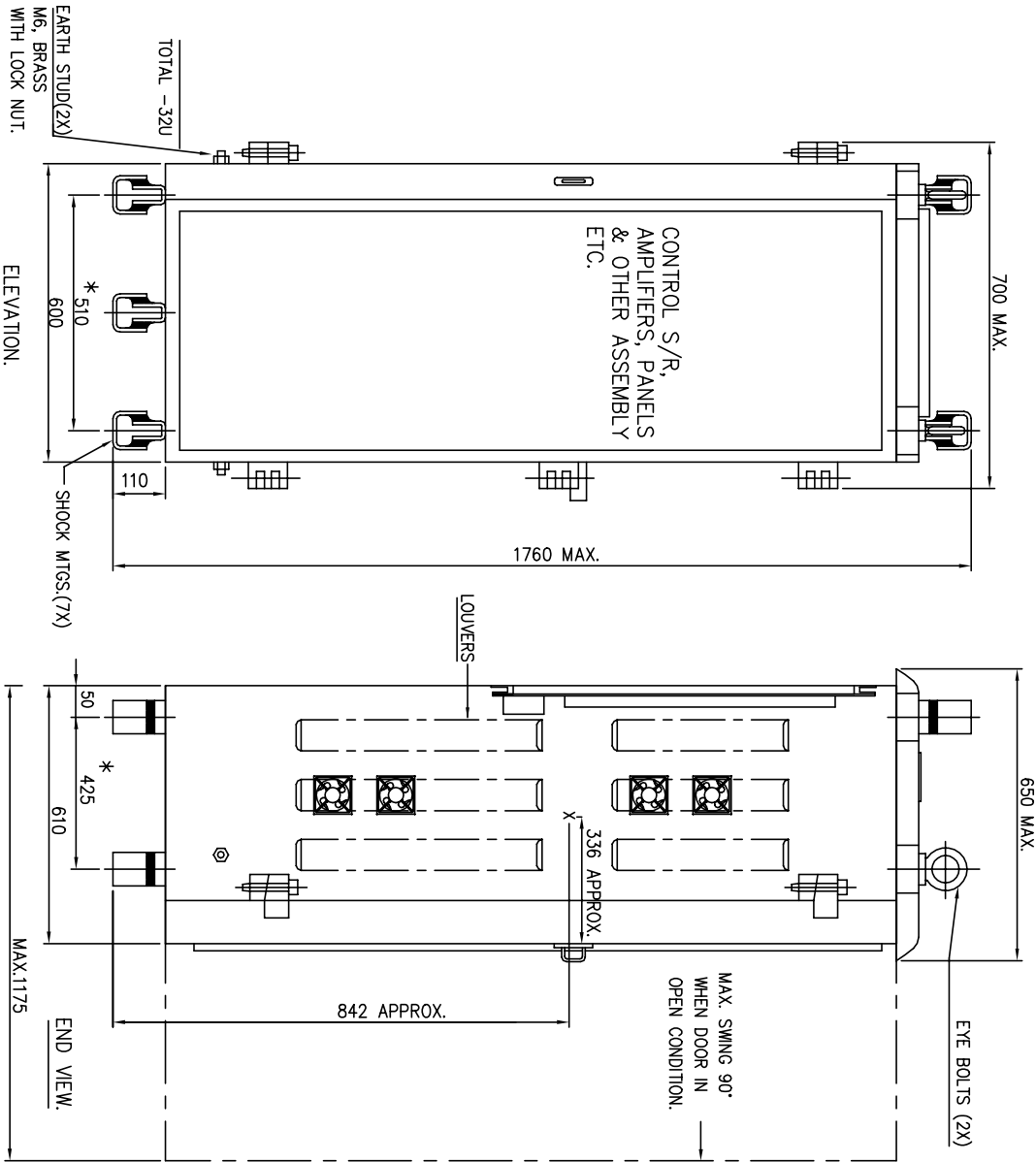
MAKE :- PHI AUDIOCOM SYSTEM PVT. LTD.
MODEL NO. :- PAN1001-1881-2
TYPE :- FLOOR MOUNTING TYPE RACK
MATERIAL OF CONSTRUCTION :- MS CRCA SHEET.
RACK FRAME - 3 mm. THICK, SWING FRAME - 2.5 mm. THICK.
COLOUR :- IS5-COLOUR CODE 632 (DARK ADMIRALTY GREY).
GLAND PLATE :- 1.6 mm THICK.
WEIGHT :- 300 kg MAX.
OPERATING VOLTAGE : 230V, 1 ϕ , 50 Hz, AC POWER SUPPLY.
POWER CONSUMPTION : 3000VA MAX.
HEAT DISSIPATION : 500W MAX.

NOTE :

- 1) ALL DIMENSIONS ARE IN MM. GEN. TOLERANCE = ± 10 mm
- 2) MINIMUM DISTANCE OF RACK FROM WALL ON FRONT & REAR SIDE SHOULD BE 700MM
- 3) ACTUAL POSITION MAY CHANGE ACCORDING TO AVAILABILITY AND SUITABILITY OF SPACE.

REF. :

1. REFER SHEET 04 FOR TECHNICAL SPECIFICATION.



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SCALE	PREPARED BY : PSD	DATE : 08/07/2021	M/S HINDUSTAN SHIPYARD LTD	
NTS	CHECKED BY : RKK	DATE : 09/07/2021	P.O.NO:- 4500001687	
	APPROVED BY : SMK	DATE : 12/07/2021	DATED :- 30.04.2021	

M/S PHI AUDIOCOM SYSTEMS PVT. LTD.		AMPLIFIER RACK ; MODEL:- PAN1001-1881-2			
G.A.	DRAWING REF. NO.: BD/P1263/MB	SHEET	03	OF	04

TECHNICAL SPECIFICATIONS

MAIN UNIT (AMPLIFIER RACK); PAN1001-1881-2

1. Mains supply voltage : 230V AC, 50Hz, single phase from UPS
2. Power consumption : 3000 VA Max @ speech load.
3. Inputs : a) ANNOUNCEMENT STATION ZONE-01,
b) ANNOUNCEMENT STATION ZONE-02,
c) MUSIC etc.
4. Outputs : High Level; For Loudspeakers
a) Zone-01
b) Zone-02

Activation to Flasher Unit
24VDC (21.6 – 26.4V)
5. Local Audio Monitoring : a) Zone-01 Output,
b) Zone-02 Output,
c) Standby Amplifier Output.
6. External connections : Through screwless Wago terminals.
7. Standby arrangement : 240W amplifier for above Zones on selective basis through Relay Panel.
8. Controls & indications : a) Mains ON indication & mains switch on mains panel,

b) Service socket for servicing purpose on base plate,

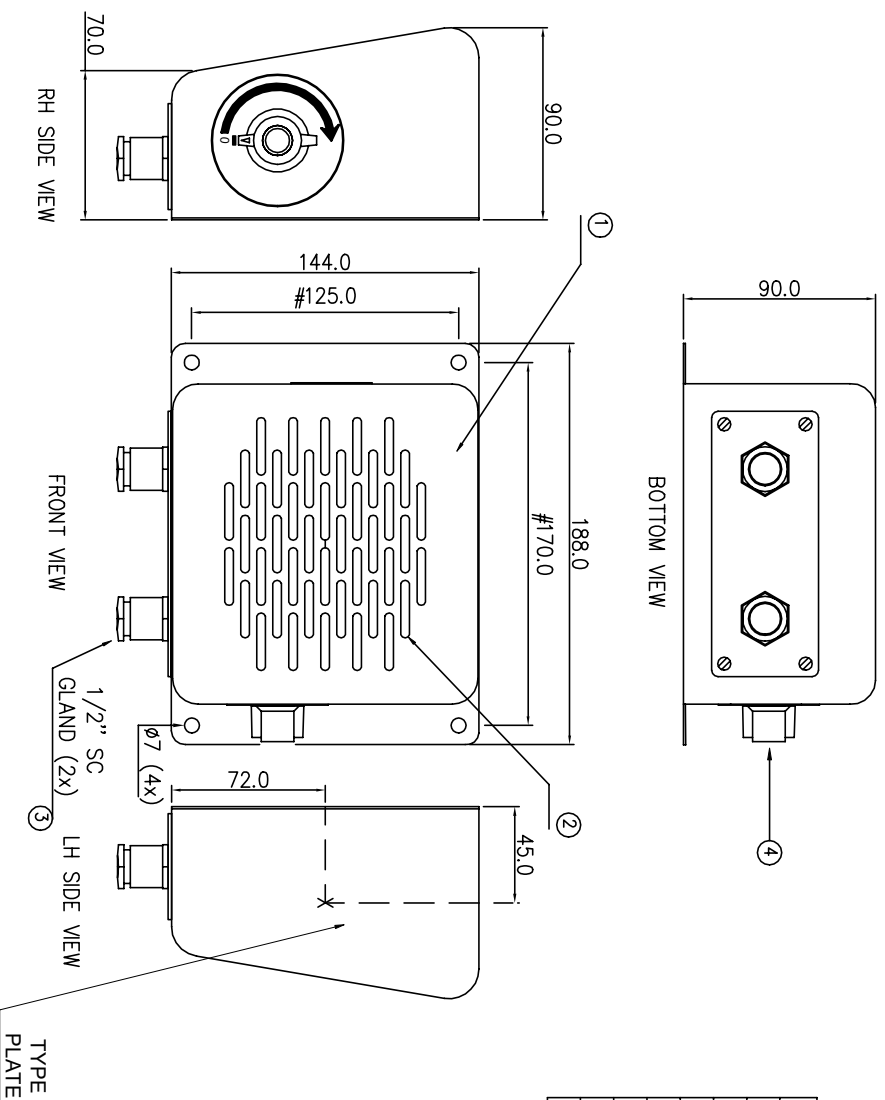
c) Switch on base plate for Ventilation fans.
9. Weight : 300 Kg. ± 20%.
10. Environmental specifications : JSS 55555 Test Nos. 10, 17, 20, 24, & 28.
11. EMI/EMC specifications : Amplifiers used are compliant with MIL STD 461-E

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ISSUE	CHANGE			DATE	APPROVED BY		
PREPARED BY: PSD		DATE: 12/07/2021		M/S: HINDUSTAN SHIPYARD LTD PO NO: 4500001687 DATED: 30.04.2021			
CHECKED BY: RKK		DATE: 13/07/2021					
APPROVED BY: SMK		DATE: 14/07/2021					
M/s PHI AUDIOCOM SYSTEMS PVT LTD		MAIN UNIT (AMPLIFIER RACK) FOR MB/SRE SYSTEM					
		DOC REF. NO.: BD/P1263/MB			SHEET	04	OF

GENERAL ARRANGEMENT DRAWING

REFERENCE TO FITTINGS:

S/N NO.	DESCRIPTION	PART NO	MATERIAL / MAKE	QTY.
1	BOX AND REAR PLATE	2065547	M.S. FABRICATED; PHI	1 NO.
2	SPEAKER 8 OHM	1201240	PHI	1 NO.
3	GLAND	2101740	BRASS NICKEL PLATED PHI	2 NO.
4	KNOB	3703261	KNOB PHI	1 NO.
5	VOLUME CNTL	3199980	ROTARY SW INTEGRAL/EOV	1 NO.
6	TERMINALS	2101980	WAGO/EOV	3 NO.
7	TRANSFORMER	2065946	COPPER PHI	2 NO.



REF: 1

1. REFER SHEET 2 FOR TECH SPECS

GENERAL TOLERANCE ± 2 MM. WEIGHT : 2.5KG MAX.

NOTE:-

- 1) # MARKED DIMENSIONS ARE MOUNTING DIMENSIONS (TOLERANCE:±1MM).
- 2) CG AS SHOWN BY 'X' APPROX.
- 3) HARDWARE USED FOR ASSEMBLY IS STANDARD STAINLESS STEEL INDUSTRIAL GRADE HARDWARE.
- 4) FINISH: IS5 – COLOUR CODE 632 (DARK ADMIRALTY GREY)
- 5) MATERIAL–M.S.

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ISSUE	CHANGE	DATE	APPRVD BY.		
SCALE	PREPARED BY : PSD	DATE : 08/07/2021	M/S HINDUSTAN SHIPYARD LTD		
NTS	CHECKED BY : RKK	DATE : 09/07/2021	P.O.NO:- 4500001687		
	APPROVED BY : SMK	DATE : 12/07/2021	DATED :- 30.04.2021		
M/S PHI AUDIOCOM SYSTEMS PVT. LTD.	BOX LOUDSPEAKER 1/2,4W (NWT) WITH VOLUME CONTROL MODEL: PAN9625				
G.A.	DRAWING REF. NO.: BD/P/1263/MB	SHEET	01	OF	02

TECHNICAL SPECIFICATIONS

1. Introduction:

PAN9625 is an Indoor Loudspeaker used for SRE/MB System. As per requirement, Wattage can be set to 1W, 2W or 4W.

Terminals for System In-Out cabling (for Loudspeaker) & for connecting in-built Speaker are provided in PAN9625. Line Matching Transformers & Volume Control is provided to control the volume of Speaker.

Volume Control interconnection arrangement is made in such a way that, for in-built Speaker, Volume Control is active for SRE & Volume Control is inactive for MB/Alarms.

2. Mechanical Specifications:

- 2.1 Enclosure & Back Cover: M.S. Fabricated
- 2.2 Glands: Brass Nickel Plated x 2 Nos for following:-
System Loudspeaker Cable-In & System Loudspeaker Cable-Out.
- 2.3 Terminals: Screw-less Wago / Equivalent
- 2.4 Finish: IS-5, Colour Code 632 (Dark Admiralty Grey)
- 2.5 Weight: 2.5Kg Max
- 2.6 Mounting: Vertical

3. Electrical Specifications:

- 3.1 Rated Input Voltage: 31.6VAC RMS
- 3.2 Wattage Taps at Primary: 1W, 2W & 4W
- 3.3 Volume Control Taps at Secondary (For SRE only): FL, 1/2, 1/4, 1/8 & 0W
(With respect to Wattage setting at Primary)
- 3.4 Frequency Range: 100Hz to 10KHz
- 3.5 Impedance of Speaker Element: 8 Ohms (± 2 Ohm)
- 3.6 Sound Pressure Level at 1W at 1 Mtr at 1KHz: 84dB (± 3 dB)
- 3.7 Maximum Power Handling capacity of Speaker Element: 6W

4. Environmental Specifications:

The unit is designed to meet JSS 55555 class N2 (Drip Proof) test nos. 9, 10, 11, 17, 20, 21, 24, 27 and 28.

5. EMI/EMC Specifications:

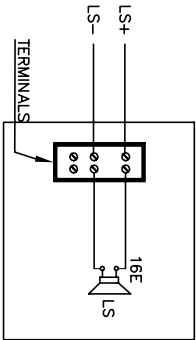
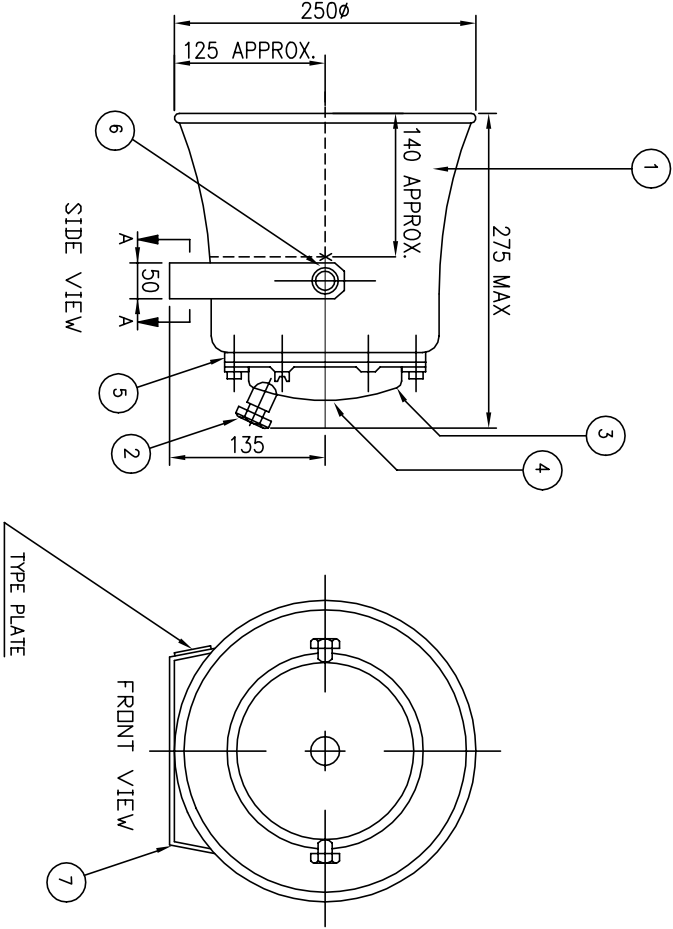
The unit involves no Electronic Parts; Hence, EMI/EMC Specifications are not applicable.

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ISSUE	CHANGE			DATE	APPROVED BY		
PREPARED BY: PSD		DATE: 12/07/2021		M/S: HINDUSTAN SHIPYARD LTD PO NO: 4500001687 DATED: 30.04.2021			
CHECKED BY: RKK		DATE: 13/07/2021					
APPROVED BY: SMK		DATE: 14/07/2021					
M/s PHI AUDIOCOM SYSTEMS PVT LTD		BOX LOUDSPEAKER 4W(NWT) WITH VOLUME CONTROL ; PAN9625					
		DOC REF. NO.: BD/P1263/MB		SHEET	02	OF	02

GENERAL ARRANGEMENT DRAWING

REFERENCE TO FITTINGS:

SR. NO.	DESCRIPTION	PART NO.	MATERIAL / MAKE	QTY.
1	HORN WITH INNER REFLECTOR	3731970	ALUMINUM LM6; PHI	1 NO.
2	19MM GLAND	2102410	BRASS NICKEL PLATED; PHI	1 NO.
3	DRIVER UNIT	3026003	ELECTRO ACOUSTIC; PHI	1 NO.
4	TERMINALS	3730850	WAGO SCREWLESS TYPE 261-301	3 NOS.
5	GASKET	3732200	NEOPRENE (IRHD 50 ±10); PHI	1 NO.
6	LOCKING GEAR	0196325	NYLON, PHI	2 NOS.
7	MOUNTING BRACKET	0196161	STAINLESS STEEL; PHI	1 NO.



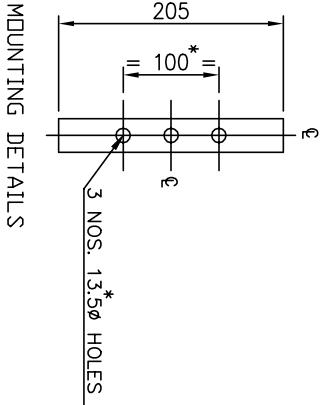
SCHEMATIC DIAGRAM

WEIGHT: 4 KG MAX

GENERAL TOLERANCE ±2 MM.

NOTE:-

- 1) REFER SHEET-2 FOR TECHNICAL SPECIFICATIONS
- 2) * MARKED DIMENSIONS ARE MOUNTING DIMENSIONS (TOLERANCE: ±1MM)
- 3) CG AS SHOWN BY 'X' APPROX.
- 4) HARDWARE USED FOR ASSLY IS STANDARD STEINLESS STEEL INDUSTRIAL GRADE HARDWARE



MOUNTING DETAILS

00	CHANGE	DATE	APPRVD BY.
SCALE	PREPARED BY : PSD	DATE : 08/07/2021	M/S HINDUSTAN SHIPYARD LTD
NTS	CHECKED BY : RKK	DATE : 09/07/2021	P.O.NO:- 4500001687
	APPROVED BY : SMK	DATE : 12/07/2021	DATED :- 30.04.2021
M/S PHI AUDIOCOM SYSTEMS PVT. LTD.			
HORN LOUDSPEAKER (15W) (WT); MODEL PAN9621			
G.A.	DRAWING REF. NO.: BD/P1263/MB	SHEET 01	OF 02

TECHNICAL SPECIFICATIONS

The unit is a wall mounting type loudspeaker intended to use in Intercom and Inter phone for Indian Naval Ship. This unit is horn loaded electro-dynamic loudspeaker assembly designed to meet water tight, shockproof and blast-proof requirements. A blast trap device fitted at the throat of driver unit to protect it from damage, which may otherwise be caused by gun blast pressure. This loudspeaker is suitable for outdoor applications. PAN9621 (Horn LS 15W-WT) when used in conjunction with PAN7409 (Volume Control Unit for Horn LS), it serves as a Horn LS for conventional SRE/MB, Intercom & Interphone System.

1. Mechanical Specifications:

- a) Make : Phi Audiocom
- b) Material : Horn – Aluminum Alloy LM6 and Bracket – S.S.
- c) Weight : 4 Kg max.
- d) Glands : 1 no. of 19mm gland is provided for cable entry.
- e) Dimensions : 250mm (H) x 275mm (D)
- f) Mounting Dimensions : 100mm
- g) Finish : IS5 – Colour Code 632 (Dark Admiralty Grey).

2. Environmental Specifications:

The unit is designed for JSS 55555 class N2 (Driving Rain) test nos. 9, 10, 12, 17, 20, 21, 24, 27 and 28.

3. EMI/EMC Specifications: Not Applicable

4. Electrical Specifications:

- a. Rated impedance : 16 ohm
- b. SPL at 1W at 1KHz at 1m : 110dB ($\pm$ 3dB)
- c. Power handling capacity : 15W (max)
- d. Effective frequency range for speech : 180Hz to 6KHz
- e. Frequency response : 400Hz to 4.5KHz

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ISSUE	CHANGE	DATE	APPROVED BY
PREPARED BY: PSD	DATE: 12/07/2021	M/S: HINDUSTAN SHIPYARD LTD PO NO: 4500001687 DATED: 30.04.2021	
CHECKED BY: RKK	DATE: 13/07/2021		
APPROVED BY: SMK	DATE: 14/07/2021		
M/s PHI AUDIOCOM SYSTEMS PVT LTD	HORN LOUDSPEAKER (15W) (WT) ; PAN9621		
	DOC REF. NO.: BD/P1263/MB	SHEET	02 OF 02

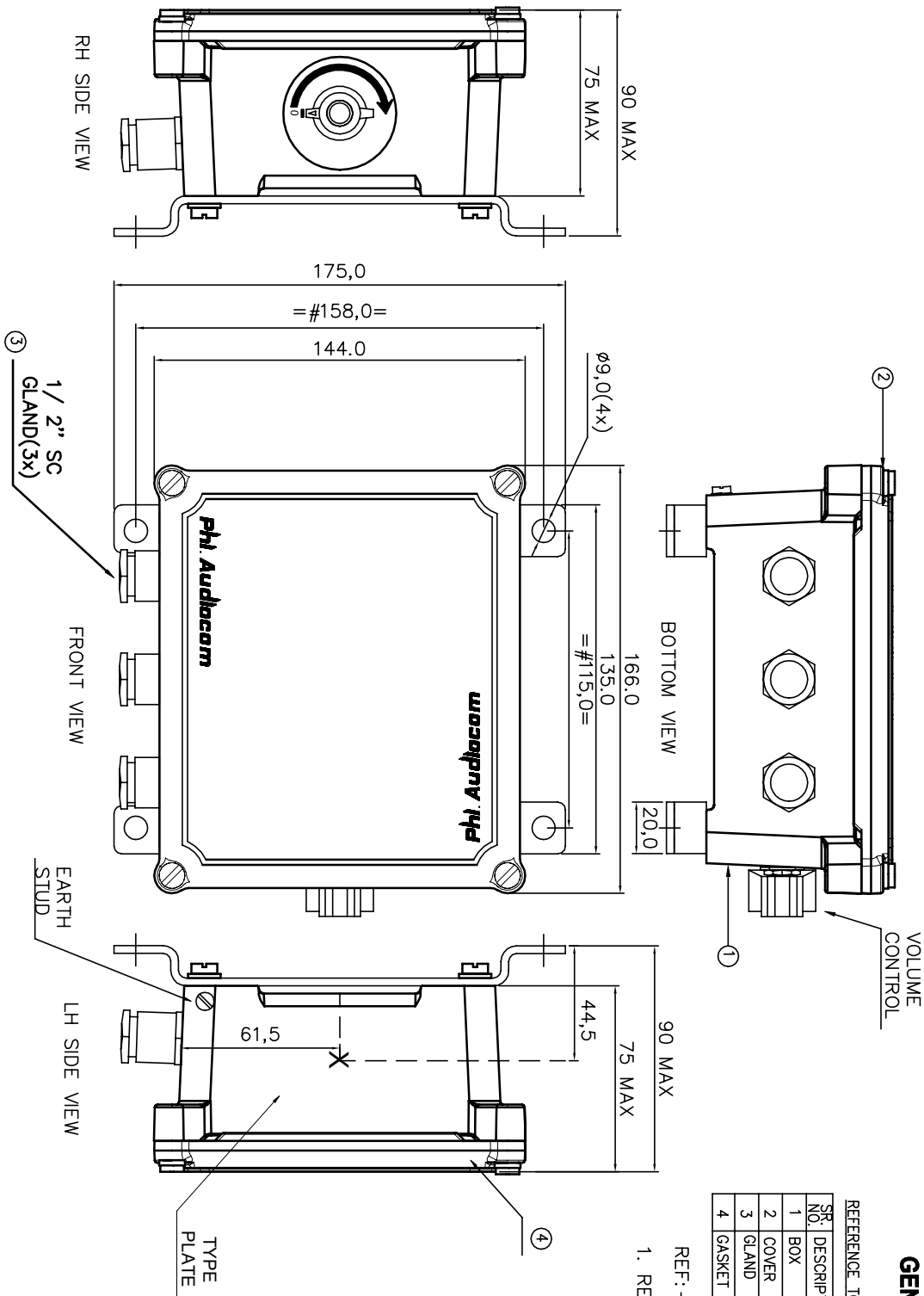
GENERAL ARRANGEMENT DRAWING

REFERENCE TO FITTINGS:

Sr. NO.	DESCRIPTION	PART NO	MATERIAL / MAKE	QTY.
1	BOX	5600860	ALUMINIUM ALLOY LM6 PHI	1 NO.
2	COVER	5600890	ALUMINIUM ALLOY LM6 PHI	1 NO.
3	GLAND	2101740	BRASS NICKEL PLATED PHI	3 NO.
4	GASKET	3543010	NEOPRENE/SILICON PHI	1 NO.

REF:-

1. REFER SHEET 2 FOR TECH SPECS



GENERAL TOLERANCE ± 2 MM. WEIGHT: 2.5KG MAX.

NOTE:-

- 1) # MARKED DIMENSIONS ARE MOUNTING DIMENSIONS (TOLERANCE: ± 1 MM).
- 2) CG AS SHOWN BY 'X' APPROX.
- 3) HARDWARE USED FOR ASSEMBLY IS STANDARD STAINLESS STEEL INDUSTRIAL GRADE HARDWARE.
- 4) FINISH: ISS - COLOUR CODE 632 (DARK ADMIRALTY GREY)
- 5) MATERIAL- ALUMINIUM ALLOY.

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ISSUE	CHANGE			DATE	APPRD BY.
SCALE NTS	PREPARED BY :	PSD	DATE :	08/07/2021	M/S HINDUSTAN SHIPYARD LTD
	CHECKED BY :	RKK	DATE :	09/07/2021	P.O.NO.: - 4500001687
	APPROVED BY :	SMK	DATE :	12/07/2021	DATED : - 30.04.2021
M/S PHI AUDIOCOM SYSTEMS PVT. LTD.			VOLUME CONTROL UNIT FOR HORN LS MODEL: PAN7409		
G.A.		DRAWING REF. NO.: BD/P/1263/MB		SHEET 01	OF 02

TECHNICAL SPECIFICATIONS

1. Introduction:

PAN7409 is a Volume Control Unit for Horn Loudspeaker used for SRE/MB, Intercom & Interphone System. This Unit is suitable for 2W to 15W Horn Loudspeaker.

Terminals for System In-Out cabling (for Loudspeaker) & for connecting Local Loudspeaker are provided in PAN7409. Line Matching Transformers & Volume Control is provided to control the volume of Loudspeaker.

Volume Control interconnection arrangement is made in such a way that, for associated overhead Local Loudspeaker, Volume Control is active for SRE & Volume Control is inactive for MB/Alarms.

2. Mechanical Specifications:

- 2.1 Box & Cover: Aluminium Alloy LM6 / LM9
- 2.2 Glands: Brass Nickel Plated x 3 Nos for following:-
System Loudspeaker Cable-In, System Loudspeaker Cable-Out & Local Loudspeaker Cable.
- 2.3 Terminals: Screw-less Wago / Equivalent
- 2.4 Finish: IS-5, Colour Code 632 (Dark Admiralty Grey)
- 2.5 Weight: 2.5Kg Max
- 2.6 Mounting: Vertical

3. Electrical Specifications:

- 3.1 Rated Input Voltage: 31.6VAC RMS
- 3.2 Wattage Taps at Primary: 2W, 4W, 8W & 15W
- 3.3 Volume Control Taps at Secondary (For SRE only): 0dB, -3dB, -6dB, -9dB & Com (With respect to Wattage setting at Primary)
- 3.4 Frequency Range: 100Hz to 8KHz

4. Environmental Specifications:

The unit meets JSS 55555 class N2 (Driving Rain) test nos. 9, 10, 12, 17, 20, 21, 24, 27 & 28

5. EMI/EMC Specifications:

The unit involves no Electronic Parts; Hence, EMI/EMC Specifications are not applicable.

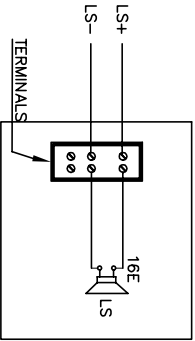
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ISSUE	CHANGE			DATE	APPROVED BY		
PREPARED BY: PSD		DATE: 12/07/2021		M/S: HINDUSTAN SHIPYARD LTD PO NO: 4500001687 DATED: 30.04.2021			
CHECKED BY: RKK		DATE: 13/07/2021					
APPROVED BY: SMK		DATE: 14/07/2021					
M/s PHI AUDIOCOM SYSTEMS PVT LTD		VOLUME CONTROL UNIT FOR HORN LOUDSPEAKER ; PAN7409					
		DOC REF. NO.: BD/P1263/MB			SHEET	2	OF

GENERAL ARRANGEMENT DRAWING

REFERENCE TO FITTINGS:

SR. NO.	DESCRIPTION	PART NO.	MATERIAL / MAKE	QTY.
1	HORN WITH INNER REFLECTOR	3731970	ALUMINUM LM6, PHI	1 NO.
2	19MM GLAND	2102410	BRASS NICKEL PLATED, PHI	1 NO.
3	DRIVER UNIT	3026003	ELECTRO ACOUSTIC, PHI	1 NO.
4	TERMINALS	3730850	WAGO SCREWLESS TYPE 261-301	3 NOS.
5	GASKET	3732200	NEOPRENE (IRHD 50 ±10); PHI	1 NO.
6	LOCKING GEAR	0196325	NYLON, PHI	2 NOS.
7	MOUNTING BRACKET	0196161	STAINLESS STEEL, PHI	1 NO.

SCHEMATIC DIAGRAM

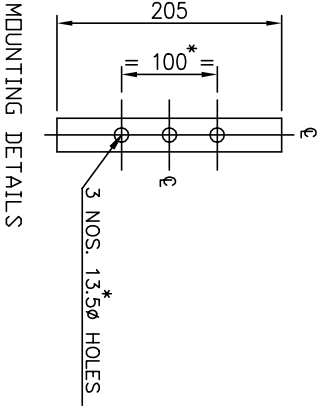
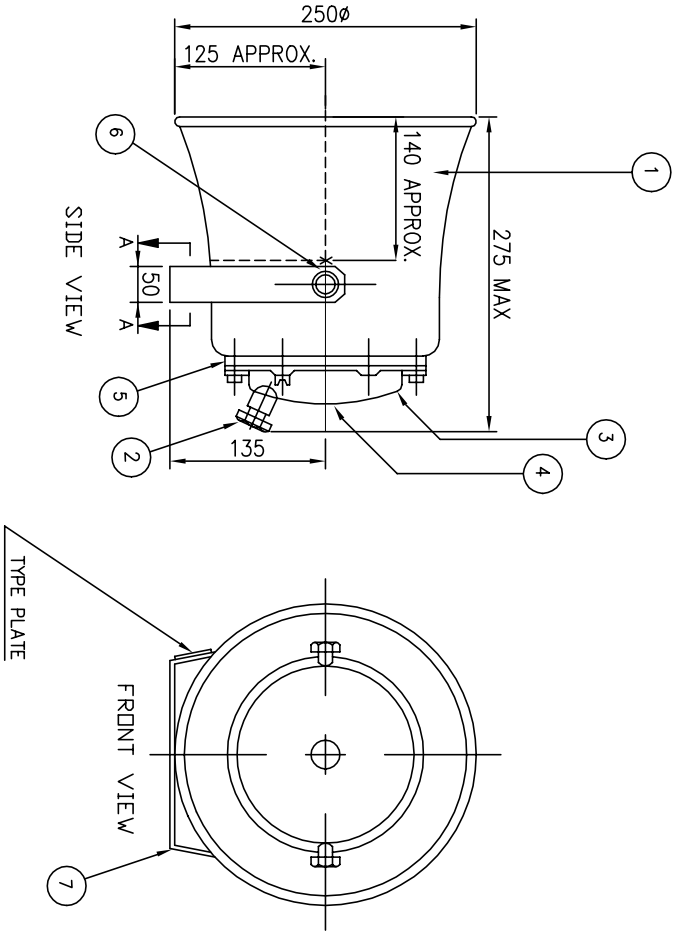


WEIGHT: 4 KG MAX

GENERAL TOLERANCE ±2 MM.

NOTE:-

- 1) REFER SHEET-2 FOR TECHNICAL SPECIFICATIONS
- 2) * MARKED DIMENSIONS ARE MOUNTING DIMENSIONS (TOLERANCE: ±1MM)
- 3) CG AS SHOWN BY 'X' APPROX.
- 4) HARDWARE USED FOR ASSLY IS STANDARD STEELNESS STEEL INDUSTRIAL GRADE HARDWARE



00	CHANGE	DATE	APPRVD BY.
ISSUE	PREPARED BY : PSD	DATE : 08/07/2021	M/S HINDUSTAN SHIPYARD LTD
SCALE	CHECKED BY : RKK	DATE : 09/07/2021	P.O.NO:- 450001687
NTS	APPROVED BY : SMK	DATE : 12/07/2021	DATED :- 30.04.2021
M/S PHI AUDIOCOM SYSTEMS PVT. LTD.			
15W (WT) HORN LOUDSPEAKER (BLASTPROOF & DUSTPROOF) ;			
PAN9621/M			
G.A.	DRAWING REF. NO.: BD/P1263/MB	SHEET 01	OF 02

TECHNICAL SPECIFICATIONS

The unit is a wall mounting type loudspeaker intended to use in Intercom and Inter phone for Indian Naval Ship. This unit is horn loaded electro-dynamic loudspeaker assembly designed to meet water tight, shockproof and blast-proof requirements. A blast trap devices fitted at the throat of driver unit to protect it from damage, which may otherwise be caused by gun blast pressure. This loudspeaker is suitable for outdoor applications. PAN9621/M (Horn LS 15W-WT) when used in conjunction with PAN7409 (Volume Control Unit for Horn LS), it serves as a Horn LS for conventional SRE/MB, Intercom & Interphone System.

1. Mechanical Specifications:

- a) Make : Phi Audiocom
- b) Material : Horn – Aluminum Alloy LM6 and Bracket – S.S.
- c) Weight : 4 Kg max.
- d) Glands : 1 no. of 19mm gland is provided for cable entry.
- e) Dimensions : 250mm (H) x 275mm (D)
- f) Mounting Dimensions : 100mm
- g) Finish : IS5 – Colour Code 632 (Dark Admiralty Grey).

2. Environmental Specifications:

The unit is designed for JSS 55555 class N2 (Driving Rain) test nos. 9, 10, 12, 17, 20, 21, 24, 27 and 28.

3. EMI/EMC Specifications: Not Applicable

4. Electrical Specifications:

- a. Rated impedance : 16 ohm
- b. SPL at 1W at 1KHz at 1m : 110dB (± 3dB)
- c. Power handling capacity : 15W (max)
- d. Effective frequency range for speech : 180Hz to 6KHz
- e. Frequency response : 400Hz to 4.5KHz

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ISSUE	CHANGE	DATE	APPROVED BY	
PREPARED BY: PSD	DATE: 12/07/2021	M/S: HINDUSTAN SHIPYARD LTD PO NO: 4500001687 DATED: 30.04.2021		
CHECKED BY: RKK	DATE: 13/07/2021			
APPROVED BY: SMK	DATE: 14/07/2021			
M/s PHI AUDIOCOM SYSTEMS PVT LTD	15W (WT) HORN LOUDSPEAKER (BLASTPROOF & DUSTPROOF) ; PAN9621/M			
	DOC REF. NO.: BD/P1263/MB	SHEET	2	OF

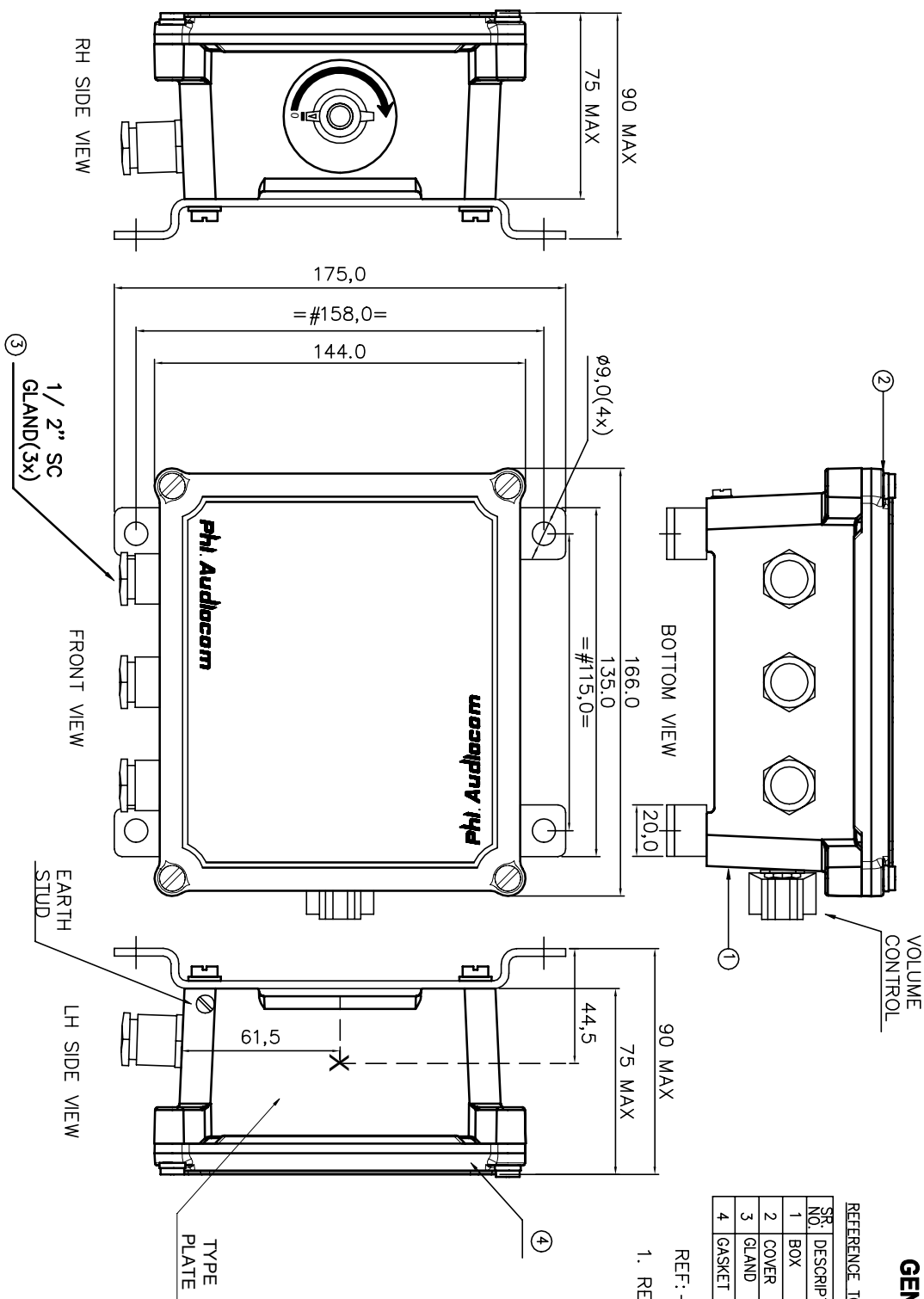
GENERAL ARRANGEMENT DRAWING

REFERENCE TO FITTINGS:

SR. NO.	DESCRIPTION	PART NO	MATERIAL / MAKE	QTY.
1	BOX	5600860	ALUMINUM ALLOY LM6 PHI 1 NO.	
2	COVER	5600890	ALUMINUM ALLOY LM6 PHI 1 NO.	
3	GLAND	2101740	BRASS NICKEL PLATED PHI 3 NO.	
4	GASKET	3543010	NEOPRENE/SILICON PHI 1 NO.	

REF:-

1. REFER SHEET 2 FOR TECH SPECS



GENERAL TOLERANCE ± 2 MM. WEIGHT 2.5KG MAX.

NOTE:-

- 1) # MARKED DIMENSIONS ARE MOUNTING DIMENSIONS (TOLERANCE: ± 1 MM).
- 2) CG AS SHOWN BY 'X' APPROX.
- 3) HARDWARE USED FOR ASSEMBLY IS STANDARD STAINLESS STEEL INDUSTRIAL GRADE HARDWARE.
- 4) FINISH: ISS - COLOUR CODE 632 (DARK ADMIRALTY GREY)
- 5) MATERIAL-ALUMINIUM ALLOY.

00	CHANGE	DATE	DATE	APPRVD BY.
ISSUE	PREPARED BY : PSD	DATE : 08/07/2021	M/S HINDUSTAN SHIPYARD LTD	
SCALE	CHECKED BY : RKK	DATE : 09/07/2021	P.O.NO.: - 4500001687	
NTS	APPROVED BY : SMK	DATE : 12/07/2021	DATED : - 30.04.2021	
M/S PHI AUDIOCOM SYSTEMS PVT. LTD.				
VOLUME CONTROL UNIT FOR HORN LS				
MODEL: PAN7409				
G.A.	DRAWING REF. NO.: BD/P1263/MB	SHEET	01	OF 02

TECHNICAL SPECIFICATIONS

1. Introduction:

PAN7409 is a Volume Control Unit for Horn Loudspeaker used for SRE/MB, Intercom & Interphone System. This Unit is suitable for 2W to 15W Horn Loudspeaker.

Terminals for System In-Out cabling (for Loudspeaker) & for connecting Local Loudspeaker are provided in PAN7409. Line Matching Transformers & Volume Control is provided to control the volume of Loudspeaker.

Volume Control interconnection arrangement is made in such a way that, for associated overhead Local Loudspeaker, Volume Control is active for SRE & Volume Control is inactive for MB/Alarms.

2. Mechanical Specifications:

- 2.1 Box & Cover: Aluminium Alloy LM6 / LM9
- 2.2 Glands: Brass Nickel Plated x 3 Nos for following:-
System Loudspeaker Cable-In, System Loudspeaker Cable-Out & Local Loudspeaker Cable.
- 2.3 Terminals: Screw-less Wago / Equivalent
- 2.4 Finish: IS-5, Colour Code 632 (Dark Admiralty Grey)
- 2.5 Weight: 2.5Kg Max
- 2.6 Mounting: Vertical

3. Electrical Specifications:

- 3.1 Rated Input Voltage: 31.6VAC RMS
- 3.2 Wattage Taps at Primary: 2W, 4W, 8W & 15W
- 3.3 Volume Control Taps at Secondary (For SRE only): 0dB, -3dB, -6dB, -9dB & Com (With respect to Wattage setting at Primary)
- 3.4 Frequency Range: 100Hz to 8KHz

4. Environmental Specifications:

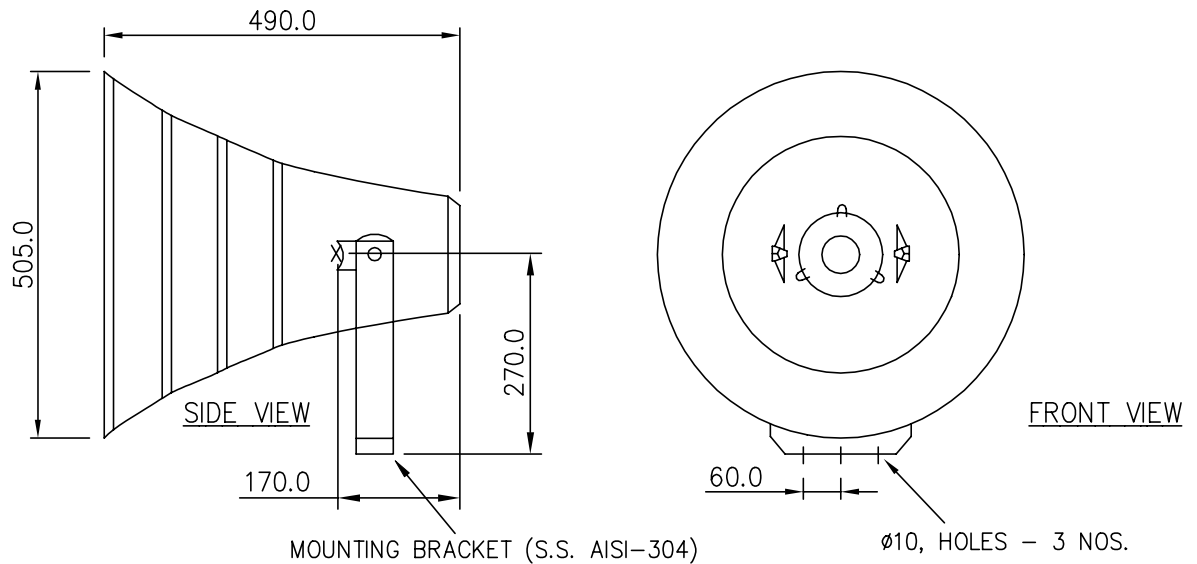
The unit meets JSS 55555 class N2 (Driving Rain) test nos. 9, 10, 12, 17, 20, 21, 24, 27 & 28

5. EMI/EMC Specifications:

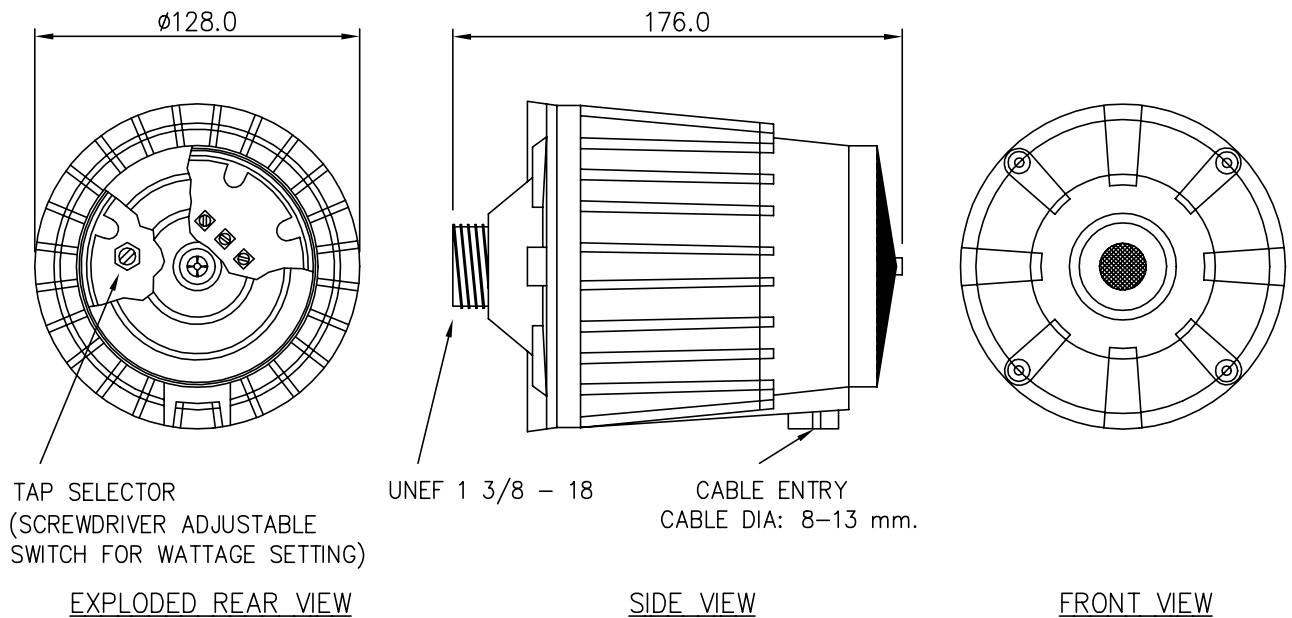
The unit involves no Electronic Parts; Hence, EMI/EMC Specifications are not applicable.

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ISSUE	CHANGE			DATE	APPROVED BY		
PREPARED BY: PSD		DATE: 12/07/2021		M/S: HINDUSTAN SHIPYARD LTD PO NO: 4500001687 DATED: 30.04.2021			
CHECKED BY: RKK		DATE: 13/07/2021					
APPROVED BY: SMK		DATE: 14/07/2021					
M/s PHI AUDIOCOM SYSTEMS PVT LTD		VOLUME CONTROL UNIT FOR HORN LOUDSPEAKER ; PAN7409					
		DOC REF. NO.: BD/P1263/MB			SHEET	2	OF

HORN LOUDSPEAKER 50W (WT):



DRIVER UNIT DST-100:



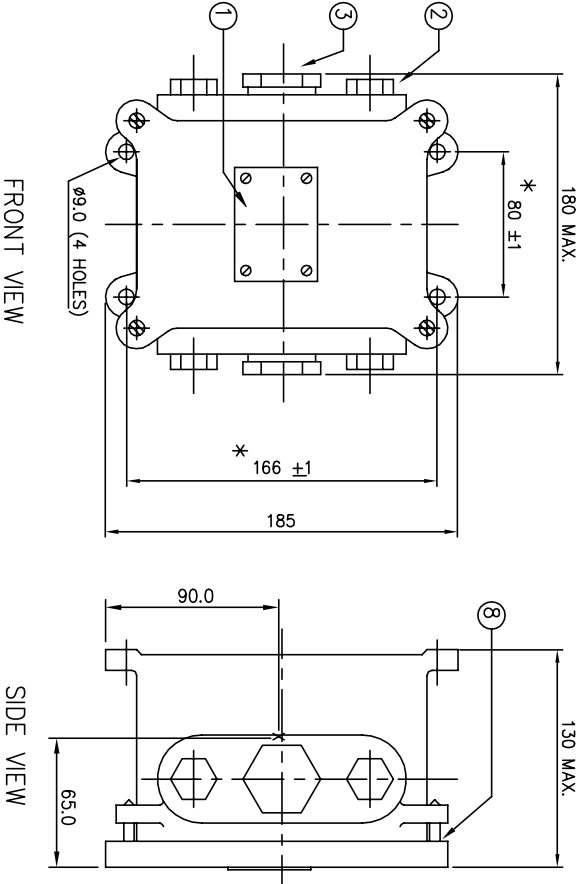
GENERAL TOLERANCE ±2 MM.

NOTE: -

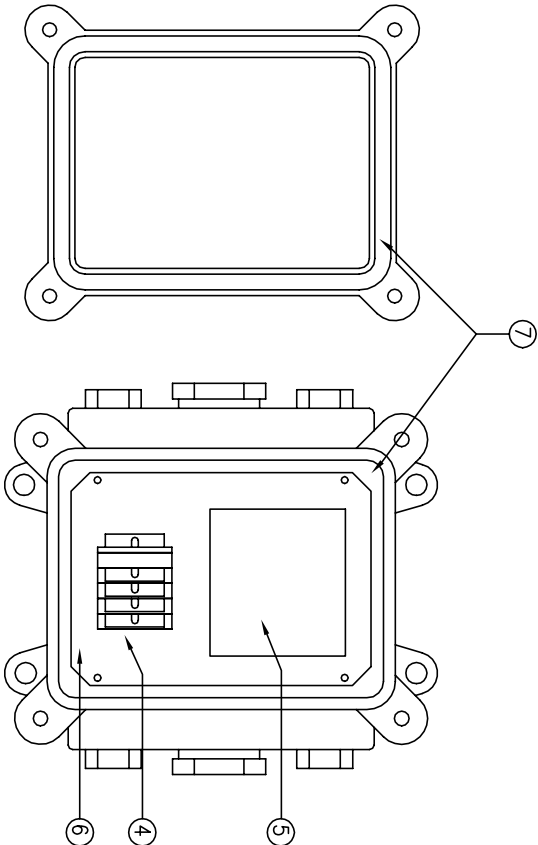
- 1) FOR G.A DRAWING & SCHEMATIC DIAGRAM OF LMT REFER SHEET NO. 02.
- 2) FOR TECHNICAL SPECIFICATIONS REFER SHEET NO. 03.
- 3) C.G. AS SHOWN BY 'X' APPROX.
- 4) HARDWARE USED FOR ASSEMBLY IS STAINLESS STEEL (AISI-304).

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ISSUE	CHANGE			DATE	APPRVD BY.	
NTS	PREPARED BY : PSD	DATE : 08/07/2021		M/S HINDUSTAN SHIPYARD LTD P.O.NO.: – 4500001687 DATED :– 30.04.2021		
	CHECKED BY : RKK	DATE : 09/07/2021				
	APPROVED BY : SMK	DATE : 12/07/2021				
M/S PHI AUDIOCOM SYSTEMS PVT. LTD.		HORN SPEAKER 50W ; MODEL DH50				
		G.A.	DRAWING REF. NO.: BD/P1263/MB		SHEET	01

LMT UNIT



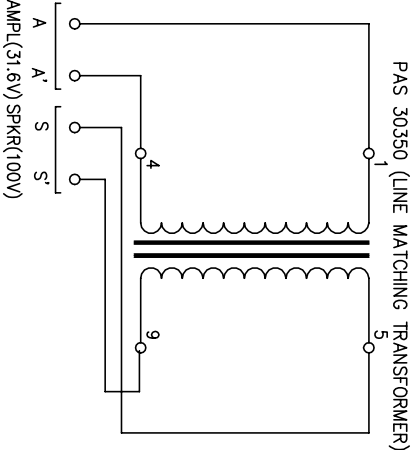
* MARKED DIMENSIONS ARE MOUNTING DIMENSIONS



REFERENCE TO FITTINGS:

SR. NO.	DESCRIPTION	PART NO.	MATERIAL / MAKE	QTY.
1	TYPE PLATE	3731650	ALUMINIUM ANODISED, PHI AUDIOCOM	1 NO.
2	GLAND NUT M20	3162340	STAINLESS STEEL (AISI-304), PHI AUDIOCOM	4 NOS.
3	GLAND NUT M36	3162370	STAINLESS STEEL (AISI-304), PHI AUDIOCOM	2 NOS.
4	TERMINALS	3730850	WAGO SCREWLESS TYPE 261-301	4 NOS.
5	TRANSFORMER	PAS 30350	LINE MATCHING TRAFO, PHI AUDIOCOM	1 NO.
6	BASE PLATE	---	M.S. (IS-513) PLATED, PHI AUDIOCOM	1 NO.
7	BOX AND COVER	3167520	ALUMINIUM LM6 PAINTED, PHI AUDIOCOM	1 NO.
8	GASKET	3700380	NEOPRENE (IRHD 50 ±10)	1 NO.

SCHEMATIC DIAGRAM



GENERAL TOLERANCE ±2 MM.

NOTE:-

- 1) FOR TECHNICAL SPECIFICATIONS REFER SHEET NO. 03
- 2) C.G. AS SHOWN BY 'X' APPROX.
- 3) HARDWARE USED FOR ASSEMBLY IS STAINLESS STEEL (AISI-304)

00	CHANGE	DATE	APPRVD BY.
ISSUE	PREPARED BY : PSD	DATE : 08/07/2021	M/S HINDUSTAN SHIPYARD LTD
SCALE	CHECKED BY : RKK	DATE : 09/07/2021	P.O.NO:- 450001687
NTS	APPROVED BY : SMK	DATE : 12/07/2021	DATED :- 30.04.2021
M/S PHI AUDIOCOM SYSTEMS PVT. LTD.			
HORN SPEAKER 50W LMT ; MODEL DH50			
G.A. DRAWING REF. NO.: BD/P1263/MB			

TECHNICAL SPECIFICATIONS

1. Mechanical specifications:

- a. The loudspeaker is used in loudhailer, Intercom system.
- b. Make: M/s DNH, Norway.
- c. Construction:
The driver horn is made in fiberglass-laminated polyester. The driver unit is made of polyester coated aluminum alloy (LM-6). The driver unit is weatherproof to IP-67.
- d. Colour: RAL 7035.
- e. Weight: 8.5 Kg ($\pm 10\%$) driver unit & horn speaker.
4 Kg ($\pm 10\%$) line matching transformer unit.
- f. The horn loudspeaker consists of three parts:
 - i. Horn loudspeaker.
 - ii. Driver unit.
 - iii. Line matching transformer unit (LMT unit).
- g. The LMT unit comprises of transformer to match the loudspeaker to 31.6 V amplifier output line. The driver unit DST-100 has rated input of 100V RMS. The LMT unit raises the 31.6 V amplifier output to 100V. The driver unit DST-100 has power selection of 50W, which can be set by means of screwdriver adjustable switch. The enclosure of LMT unit is of watertight category. The LMT unit is mounted close to the horn loudspeaker. Flexible cable (3 core, 0.75 sq. mm) of 5 meters length is supplied to interconnect the horn loudspeaker to the LMT unit.

2. Environmental specifications:

The horn loudspeaker is weatherproof to IP-67. Operating temperature is + 200 deg C / -50 deg C.

3. Electrical specifications (Driver unit DST-100):

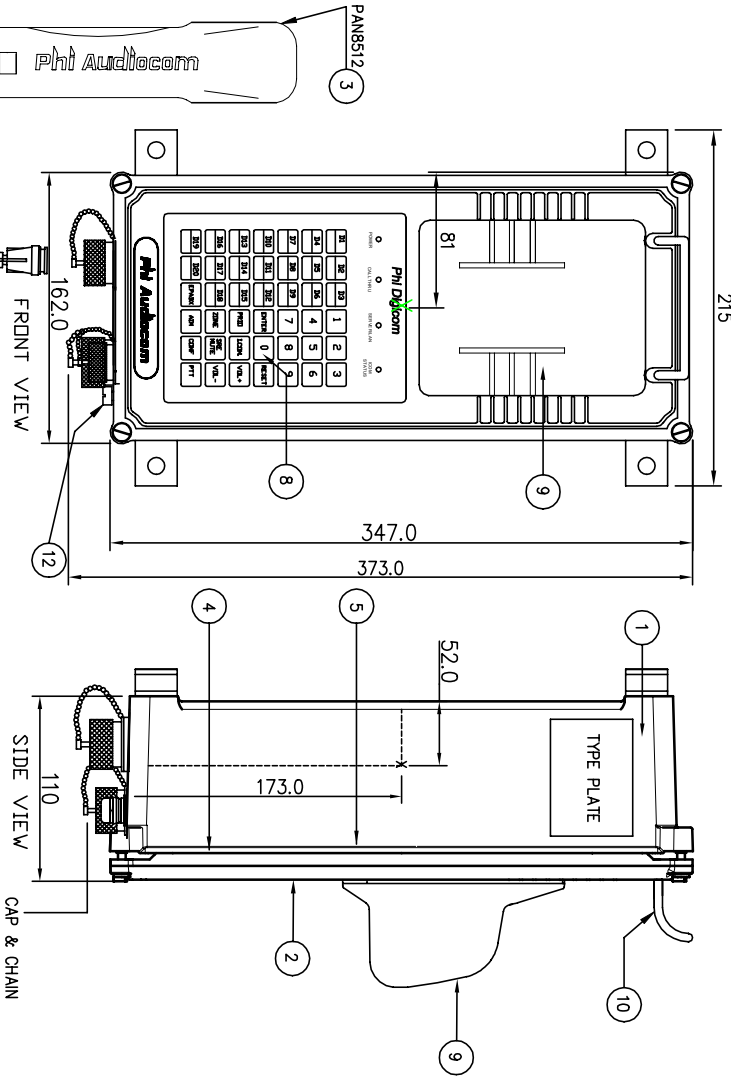
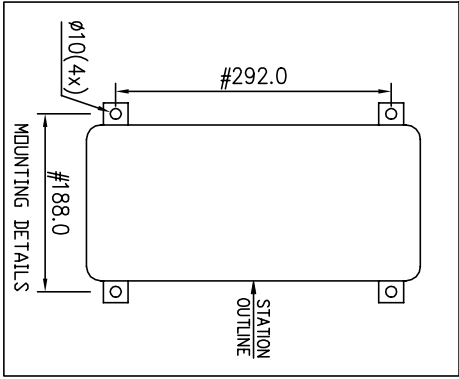
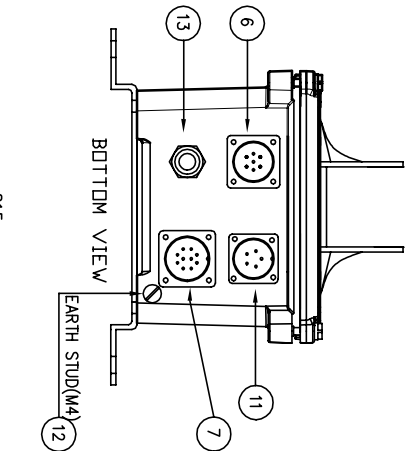
- a. Rated input voltage : 100V AC (RMS).
- b. Effective frequency range : 200 Hz to 5000 Hz.
- c. SPL at 1W at 1 KHz at 1mtr. : 108dB (± 3 dB).
- d. Power handling capacity : 50 Watts.

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ISSUE	CHANGE	DATE	APPROVED BY
PREPARED BY: PSD	DATE: 12/07/2021	M/S: HINDUSTAN SHIPYARD LTD PO NO: 4500001687 DATED: 30.04.2021	
CHECKED BY: RKK	DATE: 13/07/2021		
APPROVED BY: SMK	DATE: 14/07/2021		
M/s PHI AUDIOCOM SYSTEMS PVT LTD	HORN SPEAKER 50W (WT) ; DH50		
	DOC REF. NO.: BD/P1263/MB	SHEET	3 OF 3

GENERAL ARRANGEMENT DRAWING OF
WALL MOUNTED HANDSET STATION ; PAN5051

REFERENCE TO FITTINGS:

SR NO.	DESCRIPTION	MATERIAL / MAKE	QTY.
1	BOX	ALUMINIUM ALLOY LMG: PHI	1 NO.
2	COVER	ABS: PHI (NOT PART OF STD. PAN5051)	1 NO.
3	HANDSET ASSEMBLY	NEOPRENE (IRHD 50 $\pm$ 10); PHI	1 NO.
4	RUBBER GASKET	CONDUCTIVE; PHI	1 NO.
5	EMC GASKET	ALUMINIUM: PHI	1 NO.
6	7 PIN SOCKET (MALE)	ALUMINIUM: PHI	1 NO.
7	10 PIN SOCKET (MALE)	ALUMINIUM: PHI	1 NO.
8	KEYBOARD	KEYBOARD, PHI AUDIOCOM	1 NO.
9	GRADE ASSEMBLY	ABS: PHI	1 NO.
10	HANDLE	STAINLESS STEEL; PHI	1 NO.
11	5 PIN SOCKET (MALE)	ALUMINIUM: PHI	1 NO.
12	EARTHING STUD	EARTH STUD WITH KNURLING CAP SET, PHI	1 NO.
13	M20 GLAND	BRASS NICKEL PLATED: PHI	1 NO.



WEIGHT: 5 KG MAX
ALL DIMENSIONS ARE IN MM. GENERAL TOLERANCE $\pm$ 5 MM.
MOUNTING DIMENSION, TOLERANCE $\pm$ 2 MM.

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ISSUE	CHANGE		DATE	APPRVD BY.
SCALE	PREPARED BY : PSD	DATE : 08/07/2021	M/S HINDUSTAN SHIPYARD LTD	
NTS	CHECKED BY : RKK	DATE : 09/07/2021	P.O.NO:- 4500001687	
	APPROVED BY : SMK	DATE : 12/07/2021	DATED :- 30.04.2021	
M/S PHI AUDIOCOM SYSTEMS PVT. LTD.		WALL MOUNTED HANDSET STATION; PAN5051		
G.A.		DRAWING REF. NO.: BD/P1263/MB	SHEET 01	OF 02

TECHNICAL SPECIFICATIONS FOR WALL MOUNTED HANDSET STATION ; PAN5051

The unit is a wall mounting intelligent communication station, intended to use in MB/SRE system for offshore and ship/Submarine environment unit is housed in an aluminum casting comprising of two parts; box and cover. Keyboard with multiple keys (10 nos. numeric keys, 20 nos. direct call keys and function keys) are provided which can be used for different functionality. A 7 pin socket (male) & M20 GLAND is provided for connecting the handset to the unit & 230VAC I/P supply. Also cap and chain is provided for 7 pin socket (male). This Station is suitable for indoor applications.

1. Mechanical Specifications:

- a) Make : Phi Audiocom
- b) Material : Aluminum Alloy LM6
- c) Weight : 5 Kg Max.
- d) Dimensions : 347mm (L) x 162mm (W) x 110mm (H)
- e) Mounting Dimensions: 188mm x 292mm
- f) Finish : Colour DA Grey (Colour Code 632).

2. Environmental Specifications:

The unit is designed to meet JSS 55555 test nos. 10, 17, 20, 24 and 28.

3. EMI/EMC Specifications:

The unit is designed to meet MIL STD 461-E

4. Electrical Specifications:

- a. Power supply voltage:
 - i. Input Supply : 230V AC (+/-10%)
 - ii. Power Consumption : 50VA max
- b. Ethernet :
 - i. Data trans-reception : 100 Mbps
 - ii. Connectivity : RJ45 Standard
- c. Transmit Section:
 - i. MIC input level : 5mV
 - ii. Rated level on the line : 450mVrms ($\pm 10\%$)

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ISSUE	CHANGE		DATE	APPROVED BY		
PREPARED BY: PSD	DATE: 12/07/2021	M/S: HINDUSTAN SHIPYARD LTD PO NO: 4500001687 DATED: 30.04.2021				
CHECKED BY: RKK	DATE: 13/07/2021					
APPROVED BY: SMK	DATE: 14/07/2021					
M/s PHI AUDIOCOM SYSTEMS PVT LTD	WALL MOUNTED HANDSET STATION ; PAN5051					
	DOC REF. NO.: BD/P1263/MB		SHEET	3	OF	3

- iii. Frequency Response : 100Hz to 8KHz (± 3 dB)
- iv. Total Harmonic Distortion : Less than 2%
- v. Signal to Noise ratio : Less than -50 dB w.r.t. rated output

d. Receiver Section (for Earpiece):

- i. Rated input level : 40mVrms ($\pm 10\%$)
- ii. Rated output level : 775mVrms (0dB) ± 100 mV
- iii. Rated load resistance : 200 ohm ($\pm 5\%$) (190 ohm to 210 ohm)
- iv. Frequency Response : 100Hz to 6.5KHz (± 3 dB)
- v. Total Harmonic Distortion : Less than 2%
- vi. Signal to Noise ratio : Less than -50 dB w.r.t. rated output

e. Receiver Section (for loudspeaker):

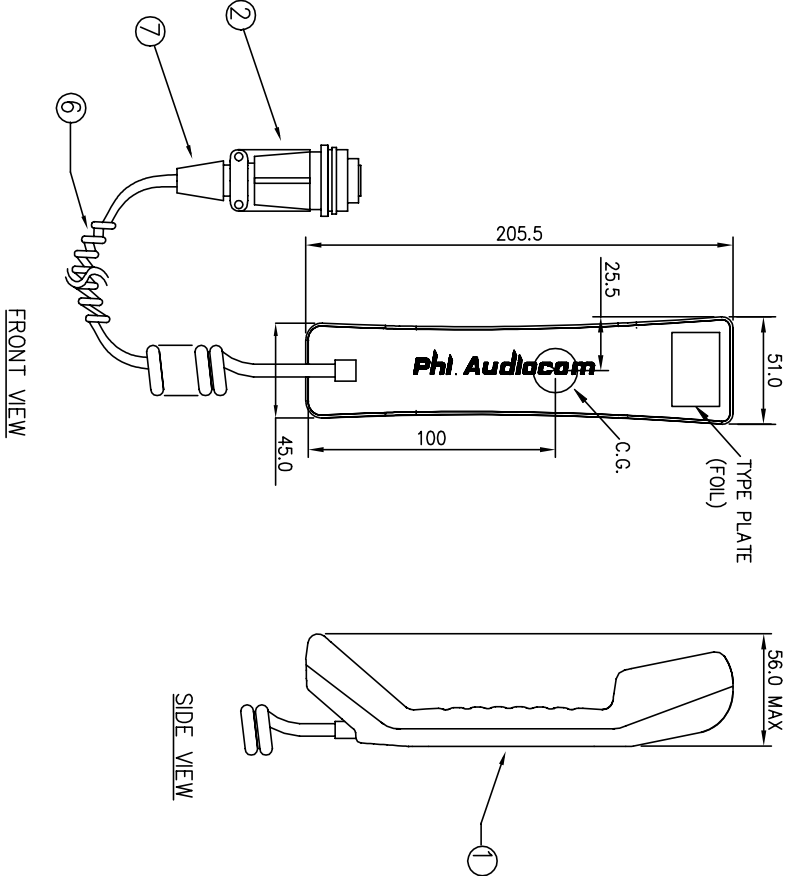
- i. Rated input level : 700mVrms ($\pm 10\%$)
- ii. Rated output level : 15.5Vrms $\pm 10\%$ Output wattage
- iii. Rated load resistance : 16 ohm ($\pm 5\%$) (15.2Vrms to 16.8Vrms)
- iv. Frequency Response : 100Hz to 8KHz (± 3 dB)
- v. Total Harmonic Distortion : Less than 2%
- vi. Signal to Noise ratio : Less than -50 dB w.r.t. rated output

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ISSUE	CHANGE	DATE	APPROVED BY	
PREPARED BY: PSD	DATE: 12/07/2021	M/S: HINDUSTAN SHIPYARD LTD PO NO: 4500001687 DATED: 30.04.2021		
CHECKED BY: RKK	DATE: 13/07/2021			
APPROVED BY: SMK	DATE: 14/07/2021			
M/s PHI AUDIOCOM SYSTEMS PVT LTD	WALL MOUNTED HANDSET STATION ; PAN5051			
	DOC REF. NO.: BD/P1263/MB	SHEET	3	OF

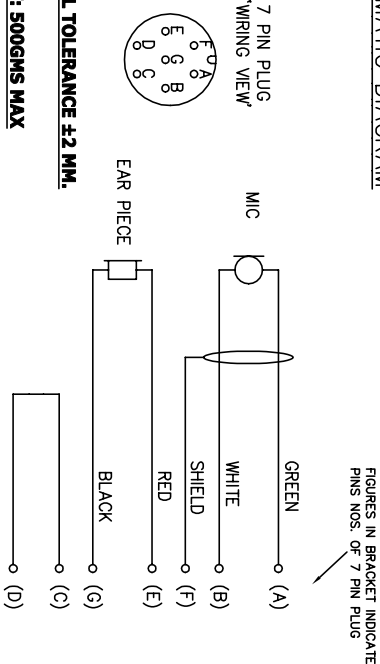
REFERENCE TO FITTINGS:

GENERAL ARRANGEMENT DRAWING

SR. NO.	DESCRIPTION	PART NO.	MATERIAL / MAKE	QTY.
1	HANDSET TOP & BOTTOM	1200460	ABS, PHI	1 NO.
2	7 PIN PLUG (FEMALE)	3718290	ALUMINIUM, AMPHENOL / ALLIED MAKE MS-3106F-16S-1S	1 NO.
3	EARPIECE	3902740	DYNAMIC, PHI	1 NO.
4	MIC CAPSULE	1300610	DYNAMIC, PHI	1 NO.
5	MAGNET	1300390	PHI AUDIOCOM	1 NO.
6	COIL CORD	3181640	6 CORE SHIELDED, PHI AUDIOCOM	1 NO.
7	SLEEVE GROMMET	3180900	NEOPRENE (IRHD 50 $\pm$ 10)	1 NO.



SCHEMATIC DIAGRAM



GENERAL TOLERANCE $\pm$ 2 MM.

WEIGHT: 500GMS MAX

NOTE:-

- 1) FOR TECHNICAL SPECIFICATIONS REFER SHEET NO. 02
- 2) HARDWARE USED FOR ASSLY IS STANDARD STEELNESS STEEL INDUSTRIAL GRADE HARDWARE
- 3) CG AS SHOWN BY 'X' APPROX.

00	CHANGE	DATE	APPRVD BY.
ISSUE	PREPARED BY : PSD	DATE : 08/07/2021	M/S HINDUSTAN SHIPYARD LTD
SCALE	CHECKED BY : RKK	DATE : 09/07/2021	P.O.NO.: - 4500001687
NTS	APPROVED BY : SMK	DATE : 12/07/2021	DATED : - 30.04.2021
M/S PHI AUDIOCOM SYSTEMS PVT. LTD.			
HANDSET WITH PLUG, MODEL PAN8512			
G.A.	DRAWING REF. NO.: BD/P1263/MB	SHEET 01	OF 02

TECHNICAL SPECIFICATIONS

1. Mechanical specifications:

- a. The handset is used in MB/SRE, Intercom system in conjunction with Wall Mount Handset station (PAN5051).
- b. The unit comprises of special noise canceling dynamic microphone capsule along with a sensitive earpiece and a heavy duty retractable coil cord which terminates in to a 7 pin circular lockable connector.
- c. Length of coil cord : 300mm $\pm$ 10mm (un-stretched).
: 3000mm $\pm$ 100mm (stretched).
- d. Colour of coil cord : Black.
- e. Handset body : ABS molded, Black matt finish.
- f. Extra provision : Small permanent magnet is provided inside the unit to operate reed switch heard in cradle.

2. Environmental Specifications:

Type tested with wall mounted handset station (PAN5051)

3. EMI/EMC Specifications:

Type tested with wall mounted handset station (PAN5051)

4. Electrical specifications:

a. Microphone

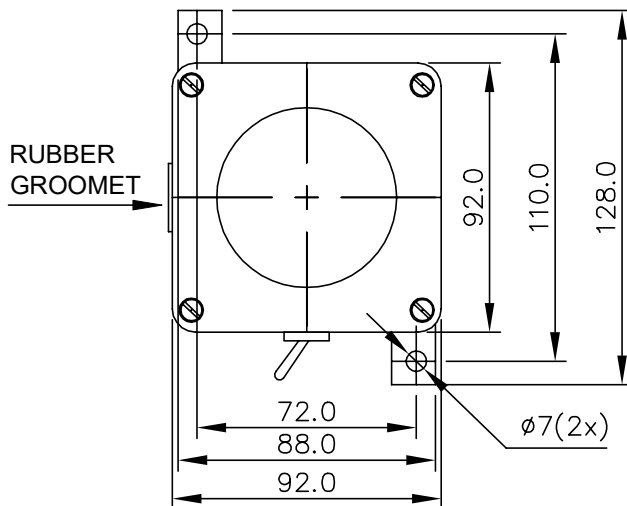
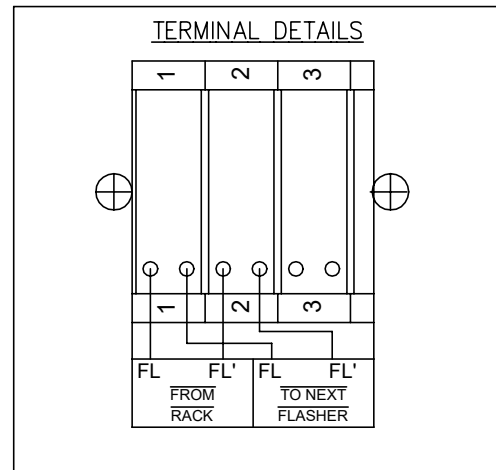
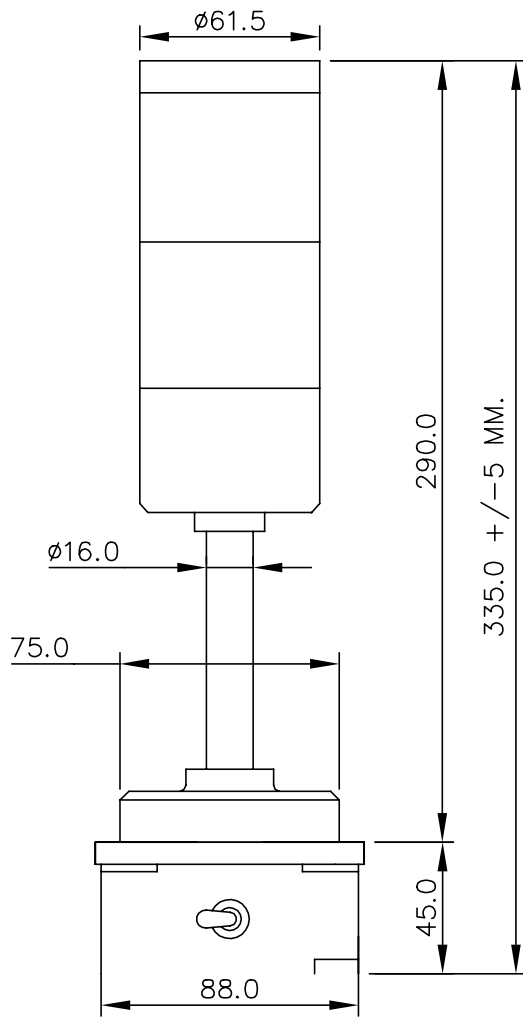
- i. Type : Noise canceling dynamic.
- ii. Rated impedance : 230 ohms at 1 KHz. $\pm$ 15 % (196 to 265 ohms).
- iii. Rated sensitivity : 0.77 mV/Pa at 1 KHz $\pm$ 3dB (0.54mV to 1.08V).

b. Earpiece

- i. Type : Dynamic.
- ii. Rated impedance : 200 ohms at 1 KHz $\pm$ 15 % (170 to 230 ohms).
- iii. Nominal SPL : 110 dB/V (Typical) at 1 KHz.
- iv. Nominal input : 3 mW.

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ISSUE	CHANGE			DATE	APPROVED BY		
PREPARED BY: PSD		DATE: 12/07/2021		M/S: HINDUSTAN SHIPYARD LTD PO NO: 4500001687 DATED: 30.04.2021			
CHECKED BY: RKK		DATE: 13/07/2021					
APPROVED BY: SMK		DATE: 14/07/2021					
M/s PHI AUDIOCOM SYSTEMS PVT LTD		HANDSET WITH PLUG ; PAN8512					
		DOC REF. NO.: BD/P1263/MB			SHEET	2	OF

GENERAL ARRANGEMENT DRAWING FOR
FLASHER ; FLASH-1881



GENERAL TOLERANCE ± 2 MM.

NOTES: -

HARDWARE USED FOR ASSEMBLY IS OF
STAINLESS STEEL (STANDARD INDUSTRIAL GRADE)

FINISH: IS5-COLOUR CODE 632 (DARK ADMIRALTY GREY)

WEIGHT: 1 KG MAX

ON/OFF SWITCH WILL BE PROVIDED ON FLASHER UNIT TO
ON/OFF AS PER THE USER REQUIREMENT

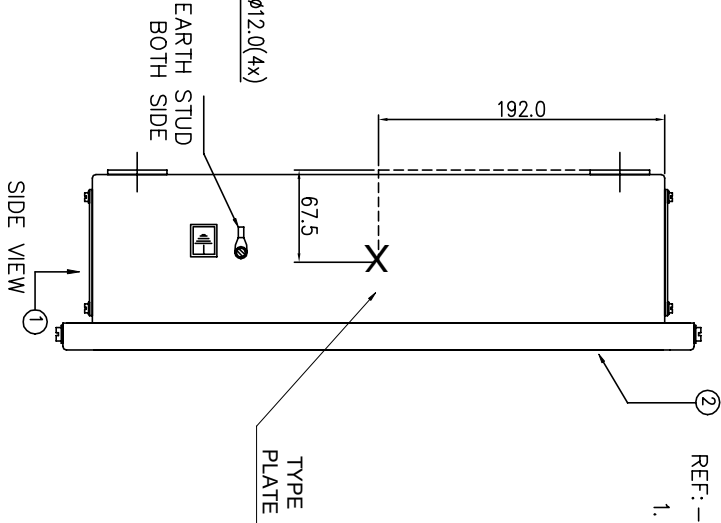
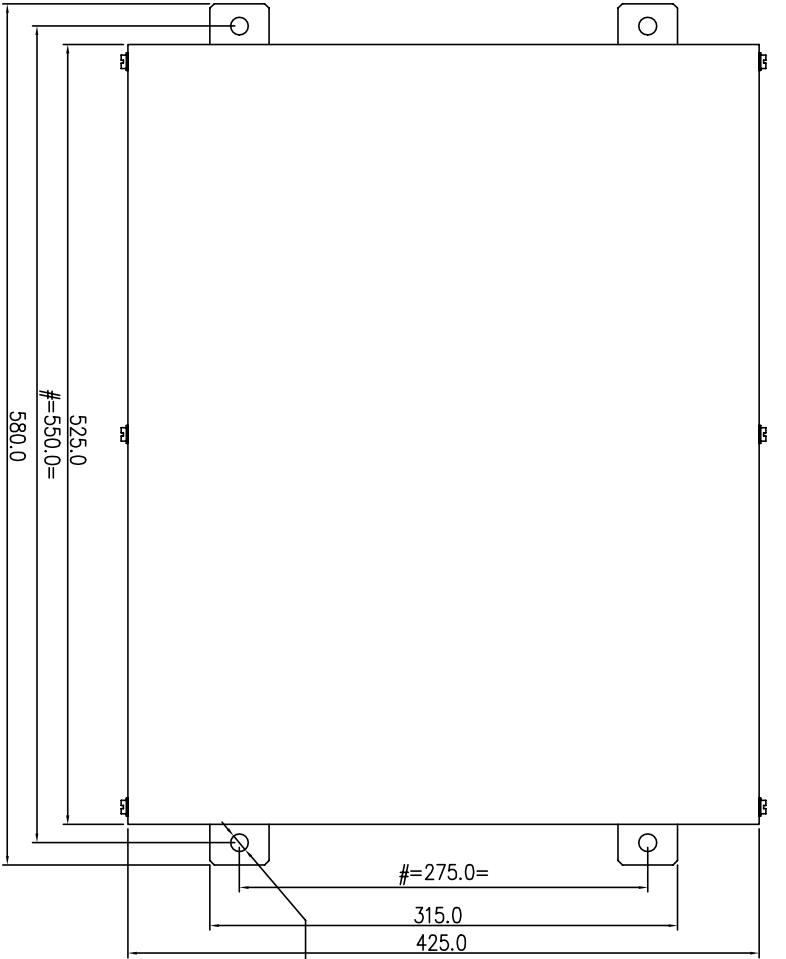
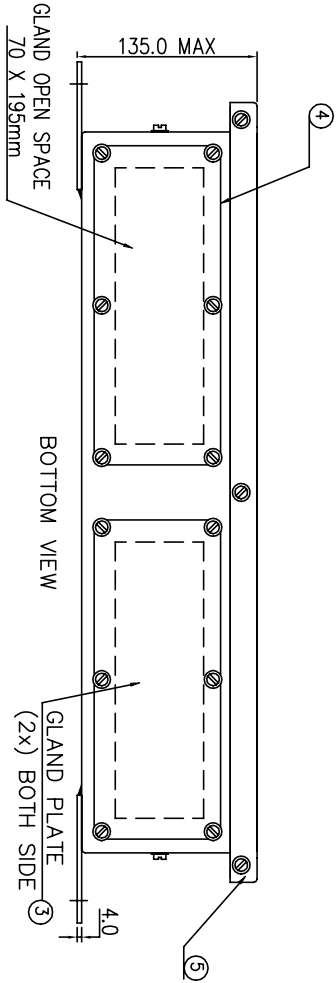
PROTECTION: IP55

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ISSUE	CHANGE			DATE	APPRVD BY.	
NTS	PREPARED BY : PSD	DATE : 08/07/2021		M/S HINDUSTAN SHIPYARD LTD P.O.NO.:– 4500001687 DATED :– 30.04.2021		
	CHECKED BY : RKK	DATE : 09/07/2021				
	APPROVED BY : SMK	DATE : 12/07/2021				
M/S PHI AUDIOCOM SYSTEMS PVT. LTD.		FLASHER; FLASH–1881				
		G.A.	DRAWING REF. NO.: BD/P1263/MB		SHEET	01

GENERAL ARRANGEMENT DRAWING

REFERENCE TO FITTINGS:

S.R. NO.	DESCRIPTION	PART NO	MATERIAL / MAKE	QTY.
1	BOX	5600820	M.S. FABRICATED: PHI	1 NO.
2	COVER	5600830	M.S. FABRICATED: PHI	1 NO.
3	GLAND PLATE	5600810	M.S. FABRICATED: PHI	4 NO.
4	GASKET GLAND PLATE	3543040	NEOPRENE/SILICON PHI	4 NO.
5	GASKET COVER	3543020	NEOPRENE/SILICON PHI	1 NO.



GENERAL TOLERANCE ± 2 MM. WEIGHT : 13KG MAX.

NOTE:-

- 1) # MARKED DIMENSIONS ARE MOUNTING DIMENSIONS (TOLERANCE: ± 1 MM).
- 2) CG AS SHOWN BY 'X' APPROX.
- 3) HARDWARE USED FOR ASSEMBLY IS STANDARD STAINLESS STEEL INDUSTRIAL GRADE HARDWARE.
- 4) FINISH: ISS - COLOUR CODE 632 (DARK ADMIRALTY GREY)
- 5) MATERIAL-M.S.

00	CHANGE	DATE	APPRVD BY.
ISSUE	PREPARED BY : PSD	DATE : 08/07/2021	M/S HINDUSTAN SHIPYARD LTD
SCALE	CHECKED BY : RKK	DATE : 09/07/2021	P.O.NO.: - 4500001687
NTS	APPROVED BY : SMK	DATE : 12/07/2021	DATED : - 30.04.2021
M/S PHI AUDIOCOM SYSTEMS PVT. LTD.			
MULTIWAY JUNCTION BOX			
MODEL: PAN2510			
G.A.	DRAWING REF. NO.: AND/P1263/MB	SHEET	01 OF 02

TECHNICAL SPECIFICATIONS

1. Introduction:

PAN2510 is a General Purpose Junction Box used for SRE/MB, Intercom & Interphone System. It is used for looping of Microphone, Loudspeaker, Power, Signal Cables, etc. The unit is a passive unit suitable for 120 wire terminations. Terminals are electrically isolated from each other.

Terminals for System In-Out cabling are provided in PAN2510.

2. Mechanical Specifications:

- 2.1 Box & Cover: M.S. Fabricated
- 2.2 Detachable Gland Plates: 4 Nos; M.S. Fabricated
- 2.3 Terminals: Screw-less Wago / Equivalent; 120 Nos (3 Columns for 40 Terminals each)
- 2.4 Finish: IS-5, Colour Code 632 (Dark Admiralty Grey)
- 2.5 Weight: 13Kg Max
- 2.6 Mounting: Horizontal

3. Environmental Specifications:

The unit is designed to meet JSS 55555 class N2 (Drip Proof) test nos. 9, 10, 11, 17, 20, 21, 24, 27 and 28.

4. EMI/EMC Specifications:

The unit involves no Electronic Parts; Hence, EMI/EMC Specifications are not applicable.

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ISSUE	CHANGE			DATE	APPROVED BY		
PREPARED BY: PSD		DATE: 12/07/2021		M/S: HINDUSTAN SHIPYARD LTD PO NO: 4500001687 DATED: 30.04.2021			
CHECKED BY: RKK		DATE: 13/07/2021					
APPROVED BY: SMK		DATE: 14/07/2021					
M/s PHI AUDIOCOM SYSTEMS PVT LTD		MULTI WAY JUNCTION BOX ; PAN2510					
		DOC REF. NO.: BD/P1263/MB			SHEET	2	OF

GENERAL ARRANGEMENT DRAWING OF NETWORK SWITCH 8 PORT WITH ENCLOSURE; SWITCH-1881

REFERENCE TO FITTINGS:

SR. NO.	DESCRIPTION	MATERIAL / MAKE	QTY.
–	COLOUR	DA GREY, CODE 632	–
1	1/2" GLAND	PHI / GS	12 NOS.
2	EARTH STUD	M4 EARTH STUD WITH KNURLING CAP SET PHI AUDIOCOM	2 NOS.

REF:-

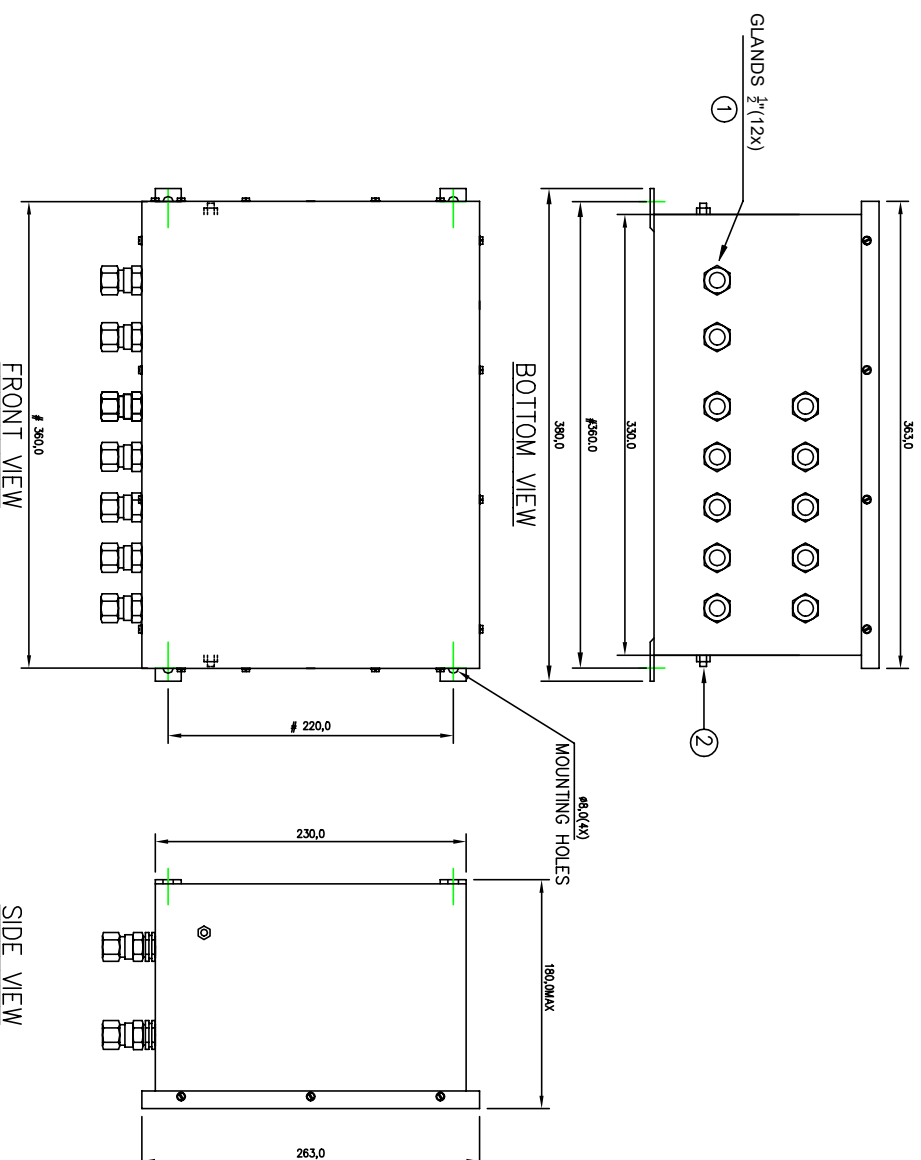
1. REFER SHEET 2 FOR TECH SPECS

WEIGHT: 11 KG APPROX

GENERAL TOLERANCE ± 5 MM.

NOTE:-

- 1) ALL DIMENSIONS IN MM.
- 2) # INDICATES MOUNTING DIMENSION.
- 3) HARDWARE USED FOR ASSEMBLY IS INDUSTRIAL GRADE STAINLESS STEEL.
- 4) # MOUNTING DIMENSION TOLERANCE IS ± 1 mm.
- 5) MATERIAL : M.S.



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ISSUE	CHANGE	DATE	APPRVD BY.
SCALE	PREPARED BY : PSD	DATE : 08/07/2021	M/S HINDUSTAN SHIPYARD LTD
NTS	CHECKED BY : RKK	DATE : 09/07/2021	P.O.NO:- 4500001687
	APPROVED BY : SMK	DATE : 12/07/2021	DATED :- 30.04.2021

M/S PHI AUDIOCOM SYSTEMS PVT. LTD.		NETWORK SWITCH 8 PORT WITH ENCLOSURE; SWITCH-1881	
G.A.	DRAWING REF. NO.: AMD/P1263/MB	SHEET 01 OF 02	

TECHNICAL SPECIFICATIONS OF NETWORK SWITCH 8 PORT WITH ENCLOSURE ; SWITCH-1881

This is wall mounting Junction Box. It is to be used for interlink between Central control unit and VOIP unit with Handset. The VOIP Junction Box is connected to Central Control Unit through FOC/ CAT6 cable & the VOIP unit with Handset is connected to VOIP JB through CAT6 Cable.

1 Mechanical Specifications:

- a) Make : Phi Audiocom
- b) Material : MS
- c) Weight : 11 Kg approx.
- d) Glands : 12 nos.
- e) Dimensions : 180mm (D) x 363mm (W) x 263mm (H)
- f) Mounting Dimensions : 360mm x 220mm
- g) Color : DA Grey (Colour CODE-632)

2 Electrical Specifications:

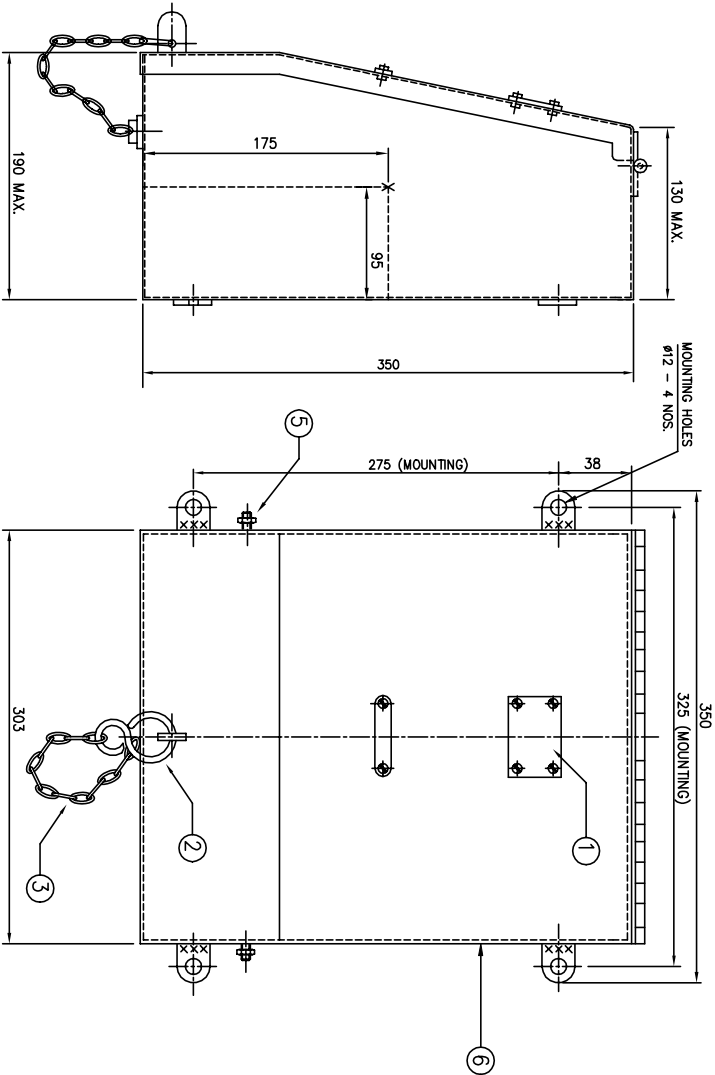
- a) Input supply : 230Vac (+/- 10%)
- b) Nos of Ethernet RJ45 ports : 8 nos
- c) VA consumption : 500VA approx

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ISSUE	CHANGE		DATE	APPROVED BY			
PREPARED BY: PSD		DATE: 12/07/2021	M/S: HINDUSTAN SHIPYARD LTD PO NO: 4500001687 DATED: 30.04.2021				
CHECKED BY: RKK		DATE: 13/07/2021					
APPROVED BY: SMK		DATE: 14/07/2021					
M/s PHI AUDIOCOM SYSTEMS PVT LTD		NETWORK SWITCH 8 PORT WITH ENCLOSURE FOR MB/SRE SYSTEM					
		DOC REF. NO.: BD/P1263/MB			SHEET	2	OF

GENERAL ARRANGEMENT DRAWING FOR
STOWAGE BOX FOR MIC; PAN1510

REFERENCE TO FITTINGS:

S.R. NO.	DESCRIPTION	MATERIAL / MAKE	QTY.
1	TYPE PLATE	ALUMINIUM ANODISED, PHI AUDIOCOM	1 NO.
2	HOOK	STAINLESS STEEL, PHI AUDIOCOM	1 NO.
3	CHAIN	BRASS, NICKEL PLATED, PHI AUDIOCOM	1 NO.
4	GRADLE FOR HANDSET	ABS : PHI	2 NOS.
5	EARTH STUD	BRASS, NICKEL PLATED, PHI AUDIOCOM	2 NOS.
6	BOX AND COVER	M.S. PAINTED, PHI AUDIOCOM	1 NO.

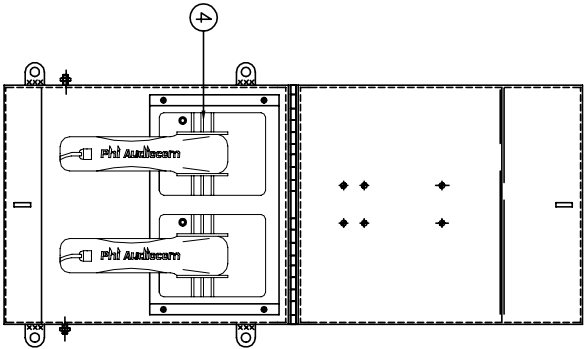


SIDE VIEW

FRONT VIEW

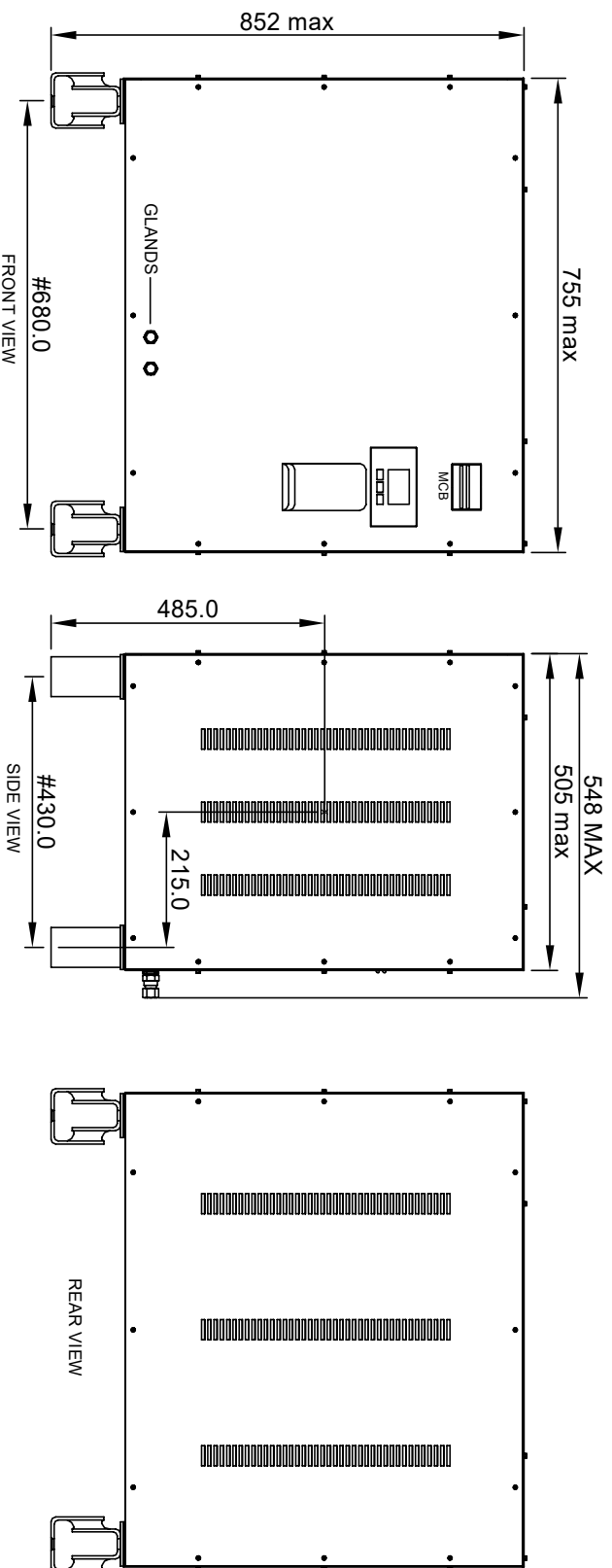
GENERAL TOLERANCE ± 5 MM. WEIGHT : 6.5 KG MAX.
NOTE:-

- 1) # MARKED DIMENSIONS ARE MOUNTING DIMENSIONS (TOLERANCE: ± 1 MM).
- 2) CG AS SHOWN BY 'X' APPROX.
- 3) HARDWARE USED FOR ASSEMBLY IS STANDARD STAINLESS STEEL INDUSTRIAL GRADE HARDWARE.
- 4) COLOUR- D.A GREY RAL 632
- 5) MATERIAL-M.S.
- 6) MAKE: PHI AUDIOCOM SYSTEMS PVT. LTD.



VIEW OF THE UNIT WITH COVER OPEN

00	CHANGE	DATE	APPRVD BY.
SCALE	PREPARED BY : PSD	DATE : 08/07/2021	M/S HINDUSTAN SHIPYARD LTD
NTS	CHECKED BY : RKK	DATE : 09/07/2021	P.O.NO:- 4500001687
	APPROVED BY : SMK	DATE : 12/07/2021	DATED :- 30.04.2021
M/S PHI AUDIOCOM SYSTEMS PVT. LTD.			
STOWAGE BOX FOR MIC; PAN1510			
G.A.	DRAWING REF. NO.: AMD/P1263/MB	SHEET 01	OF 01



NOTES:-

- 1) UPS: 6KVA
- 2) SPACE FOR BATTERY: SMF 42AH / 12V X 16 NOS
- 3) BATTERIES WILL BE SUPPLIED SEPARATELY.
- 4) GENERAL TOLERANCE ± 5 MM
- 5) WEIGHT : 300 KG MAX (WITH BATTERIES).
- 6) # MARKED DIMENSIONS ARE MNT DIMENSIONS (TOLERANCE: ± 2 MM).
- 7) CG AS SHOWN BY 'X' APPROX.
- 8) HARDWARE USED FOR ASSEMBLY IS STANDARD STAINLESS STEEL INDUSTRIAL GRADE HARDWARE.
- 9) FINISH: COLOUR RAL 632 DA GREY
- 10) MATERIAL-M.S.
- 11) GLANDS: BRASS NICKEL PLATED X 2 NOS
- 12) MCB: 25/32 AMP, SCHNEIDER OR EQUIVALENT X 2 NOS
- 13) HEAT DISSIPATION: 1000W MAX

14) Additionally, Legrand Make (Model 507613) Mains Distribution Box with MCBs (6AX 4 NOS; For CCU, For Communication Stations, Power Supply Junction Box. 32AX INO- for Amplifier rack) will be supplied.

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ISSUE	CHANGE	DATE	APPRVD BY.
SCALE	PREPARED BY : PSD	DATE : 08/07/2021	M/S HINDUSTAN SHIPYARD LTD
NTS	CHECKED BY : RKK	DATE : 08/07/2021	P.O.NO:- 4500001687
	APPROVED BY : SMK	DATE : 12/07/2021	DATED :- 30.04.2021
M/S PHI AUDIOCOM SYSTEMS PVT. LTD.			
UPS; MODEL:- UPS-1881			
G.A.			
DRAWING REF. NO.: BDP1263MB			
SHEET 01 OF 03			

Details of UPS & Batteries; UPS-1881 & BATT-1881

UPS WITH REQUIRED BATTERIES

Details of 6 KVA UPS are as follows:-

Make: Vertiv

Model: LIBERT GXT MT+ LB 6KVA UPS SYSTEM with SMF Batteries

Description: Liebert GXT MT+ LB Series systems is true double conversion online UPS systems designed to provide with a capacity of 6 KVA. Liebert GXT MT+ LB units feature total isolation of the load from the mains - isolating input and output sections. Input Supply: 230VAC, 50Hz, 1 Phase. Output Supply: 230VAC, 50Hz, 1 Phase.

VA Rating details of MB/SRE & Intercom System are as follows:

Description	Speech Load	Rated Load
MAIN UNIT; PAN1001-1881	3000 VA	3500VA
AUXILIARY UNITS (FIELD STATION & N/W SWITCH JBS)	800 VA	1100 VA
Total VA	3800 VA	4600 VA

Considering a buffer for additional Rated Load, 6KVA UPS will be supplied.

The UPS will enable satisfactory operation of above Naval Communication System for Speech Load for 1 hour in absence of 230 V AC Supply. For this UPS, the Battery Calculations are as follows:

Considering a buffer for additional Speech Load, approx 4000 VA speech load is considered.

Output KVA : 4.8 KVA
Load Power factor : 0.8
Nominal DC Voltage : 192 V
End Cell Voltage : 10.5 V (for one cell)
Inverter efficiency : 90%
No. of batteries : 16 Nos.
Backup : 1 Hrs.
Capacity Utilization : 0.62% (for 1 Hr)

DC Current = UPS (KVA) x 1000 x Load Power Factor/(Inverter efficiency x End Voltage)

$$= 4.8 \times 1000 \times 0.8 / (0.90 \times 10.5 \times 16)$$
$$= 25.4 \text{ Amp.}$$

Battery 'AH' = DC Current x Duration in Hr. / % Capacity Utilization
= 25.4 x 1 / 0.62
= 41 AH

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CHECKED BY: RKK		DATE: 13/07/2021					
APPROVED BY: SMK		DATE: 14/07/2021					
M/s PHI AUDIOCOM SYSTEMS PVT LTD		UPS WITH REQUIRED BATTERIES FOR 1 HRS BACKUP FOR MB/SRE SYSTEM					
		DOC REF. NO.: BD/P1263/MB			SHEET	2	OF

Hence; 42 AH / 12 V/ 16 Nos. SMF Batteries (Make:- Exide, Amara Raja or equivalent) will be supplied to enable satisfactory operation of above Naval Communication System for Speech Load for 1Hr in absence of 230 V AC Supply. Additionally, Legrand Make (Model 507613) Mains Distribution Box with MCBs (6A & 32A; For Central Control Unit & Amplifier rack, 6A*3Nos; For n/w switch JBs & Stations) will be supplied. 2 Nos 25A/32A MCB will be provided with UPS as I/P & O/P isolator.

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M/s PHI AUDIOCOM SYSTEMS PVT LTD		UPS WITH REQUIRED BATTERIES FOR 1 HRS BACKUP FOR MB/SRE SYSTEM					
		DOC REF. NO.: BD/P1263/MB		SHEET	3	OF	03